CSE 311:
Data Communication

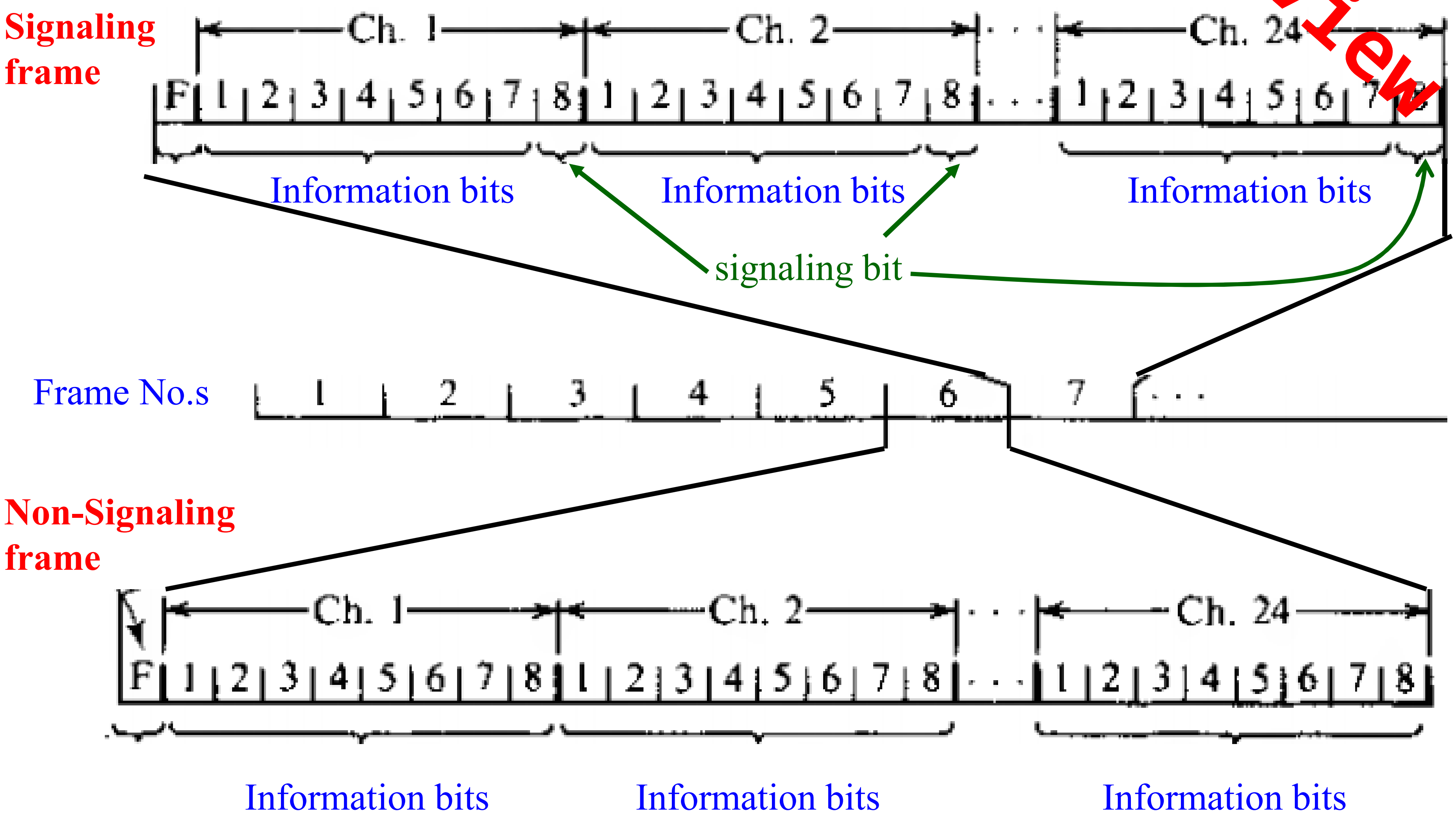
Instructor:

Dr. Md. Monirul Islam

Sampling
and
Analog-to-Digital Conversion

How to Differentiate Signaling and Non-Signaling Frames?

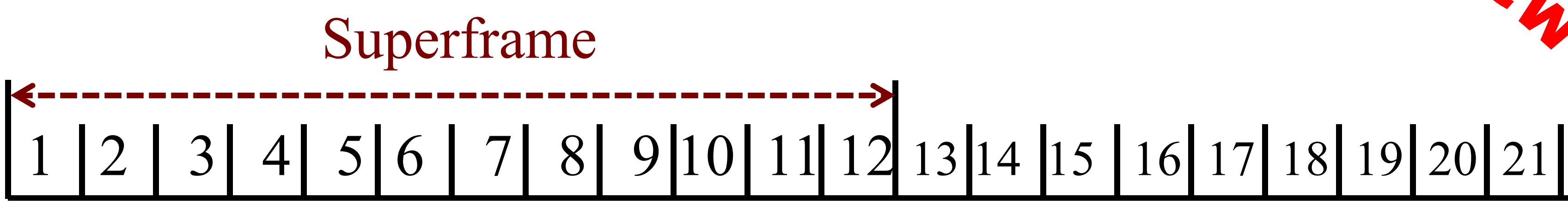
Review



How to Differentiate Signaling and Non-Signaling Frames?

Review

Concept of Superframe



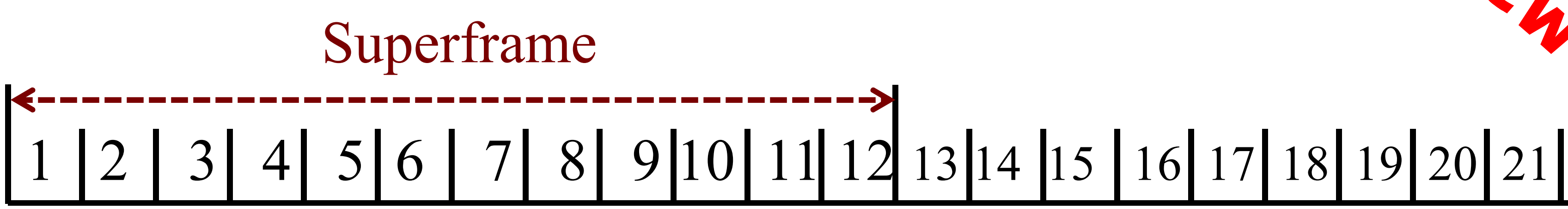
Frame No.s

- 12 frames make a Superframe
- Framing bits form a special pattern 100111011100

How to Differentiate Signaling and Non-Signaling Frames?

Review

Concept of Superframe



Frame No.s

- The special pattern 100111011100 repeats in every 12 frames
- Identification of 6th and 12th frame is easy, and so is signaling frames.

Extended Superframe (ESF)

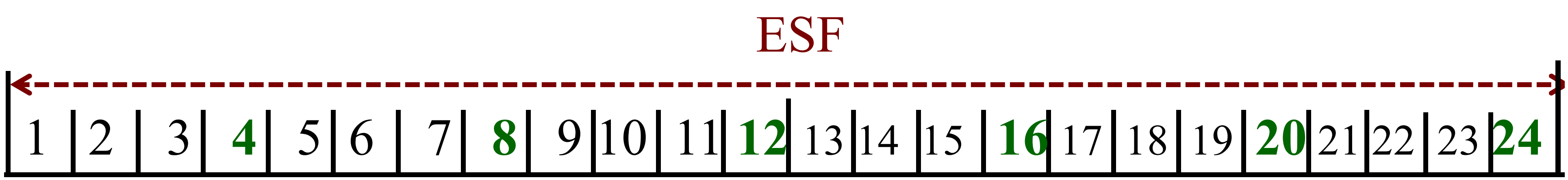
Structure of ESF

Frame No.s

- 24 frames make an ESF
- Reduces framing bits
- Signaling bits in the LSB of 6th, 12th, 18th and 24th frames
- 4 signaling bits makes possibility of 16 state-signaling

Extended Superframe (ESF)

Structure of ESF

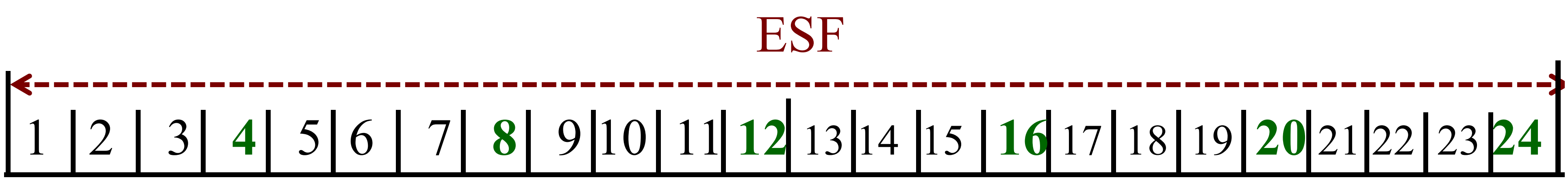


Frame No.s

- Reduces framing bits: every 4th frame has a framing bit
- 8 Kbps framing capacity is distributed to
  - 2 Kbps for framing
  - 2 Kbps for CRC-6
  - 4 Kbps for data

Extended Superframe (ESF)

Structure of ESF



Frame No.s

- Reduces framing bits: every 4th frame has a framing bit
  framing bits make a pattern **001011**: repeats in every 24 frames

Further improvement: Common Channel Interoffice Signaling

CCIS

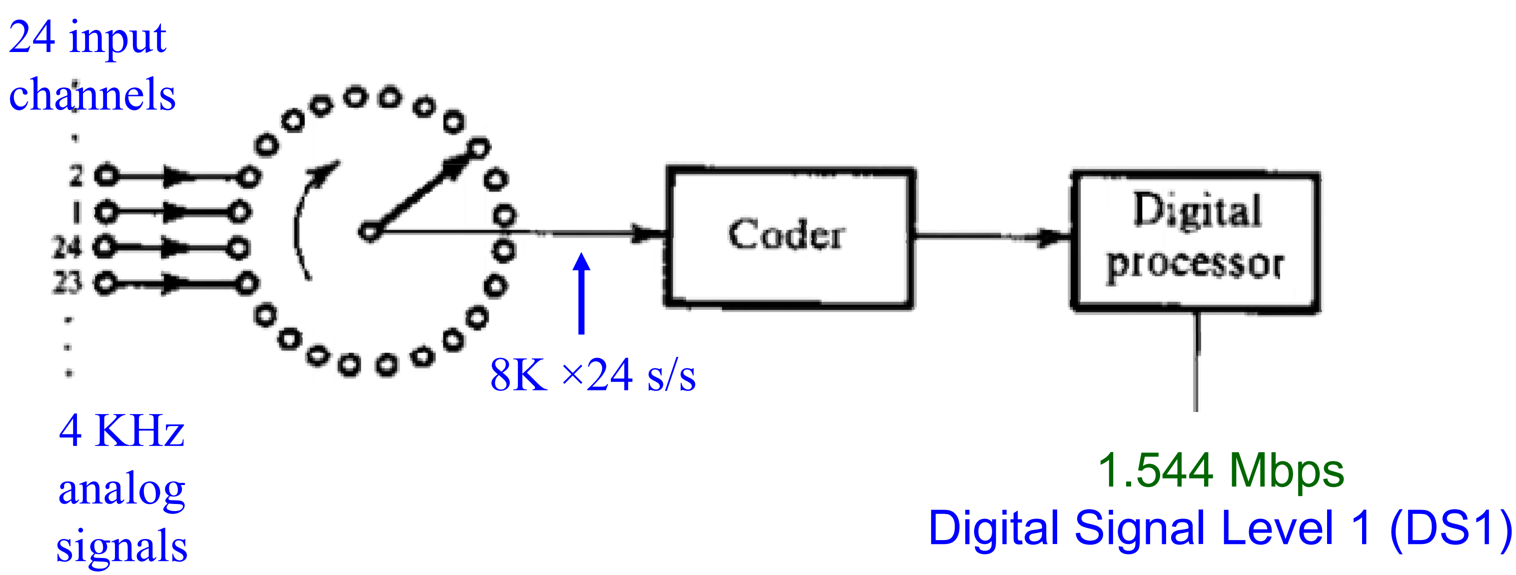
- A separate network for signaling
- decreases the necessity of robbed-bit-signaling
- All 8 bits can be used for transmitting information bits

European Counterpart of T1 System

Conference on European Postal and Telegraph Administration (CEPT)

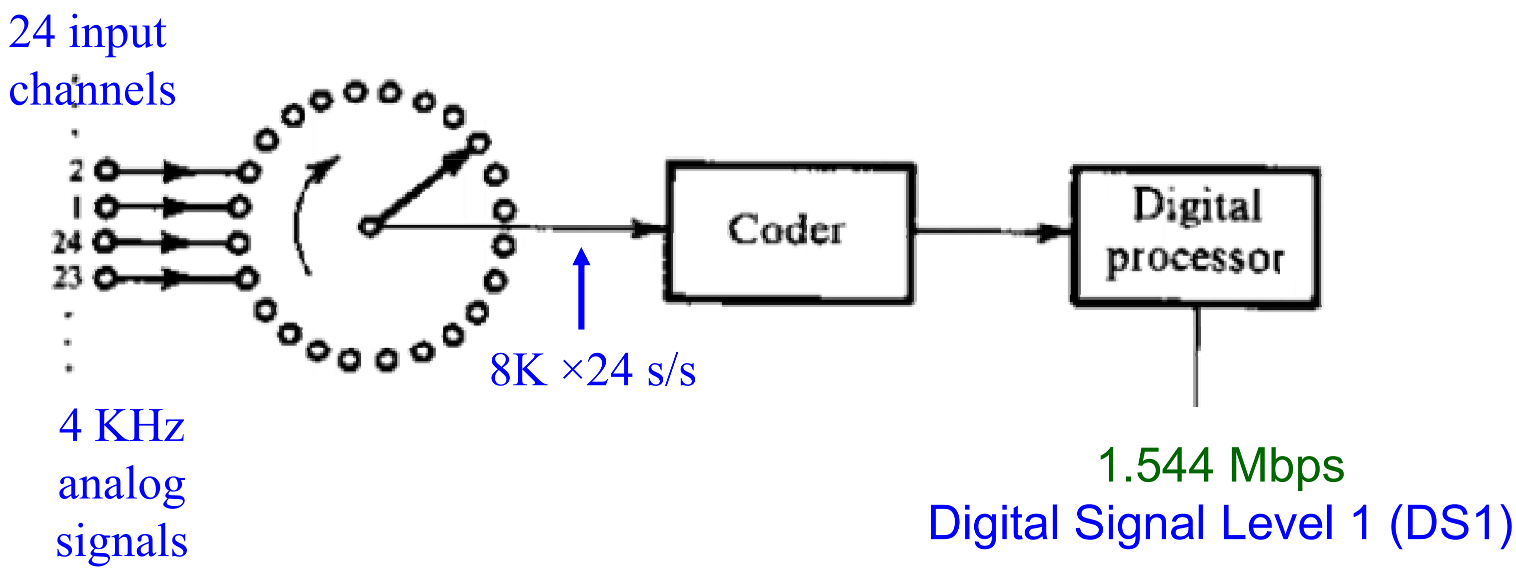
- standardizes PCM with 256 time slots per frame
- capable to transmit 32 channel audio signals simultaneously
- uses 30 channels with $30 \times 8 = 240$ information bits per frame
- uses remaining 16 bits for sync and signaling

Digital Multiplexing: Generalization of T1 System



- Converts 24 4-KHz analog signals to digital signal of 1.544 Mbps

Digital Multiplexing: Generalization of T1 System



- Several slow-bit-rate digital signals can be combined to get high-bit-rate signal and can time-share same medium (TDM)

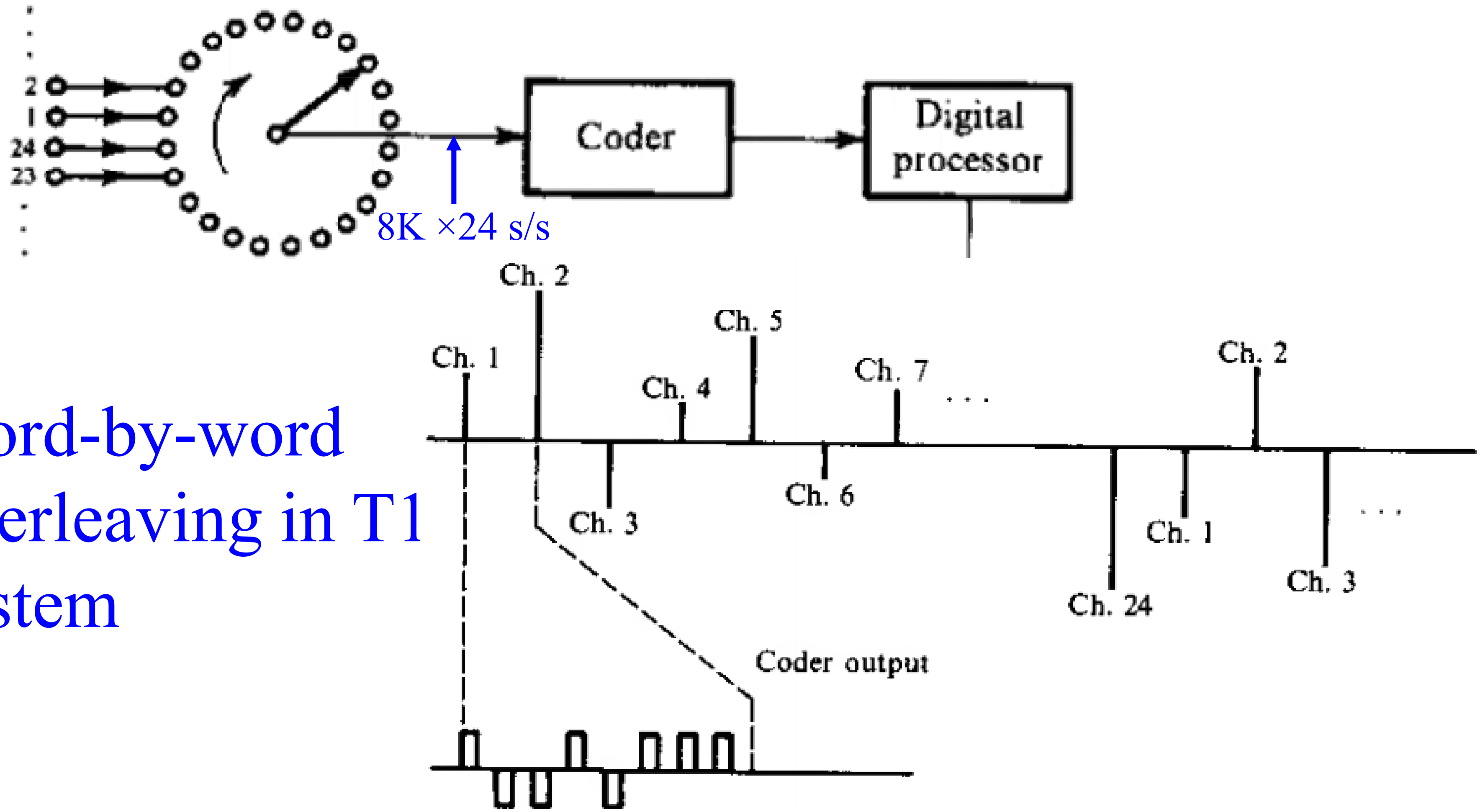
- Input bit rates may be different

Digital Multiplexing: Generalization of T1 System

TDM can be

- Word-by-word basis interleaving (like T1 system)
- Bit-by-bit basis interleaving

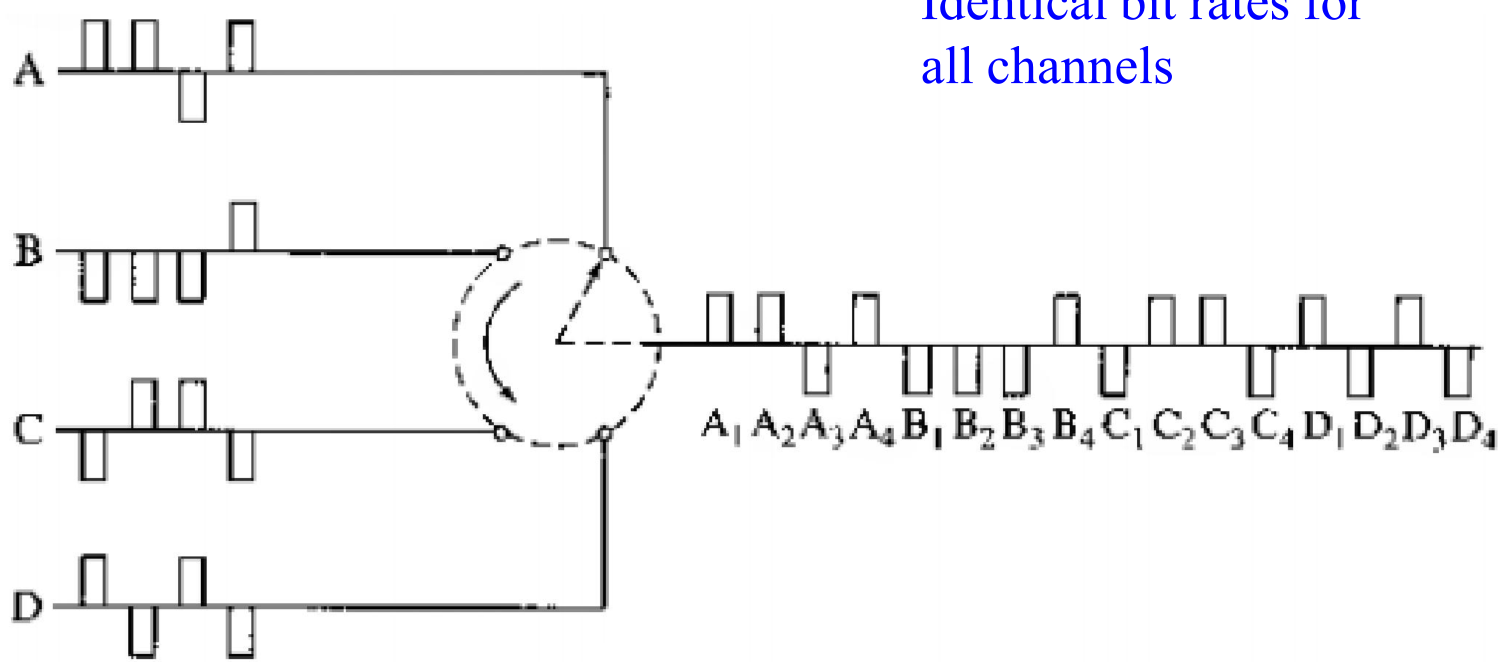
Digital Multiplexing: Generalization of T1 System



Word-by-word interleaving in T1 system

Digital Multiplexing: Word Interleaving

Channels

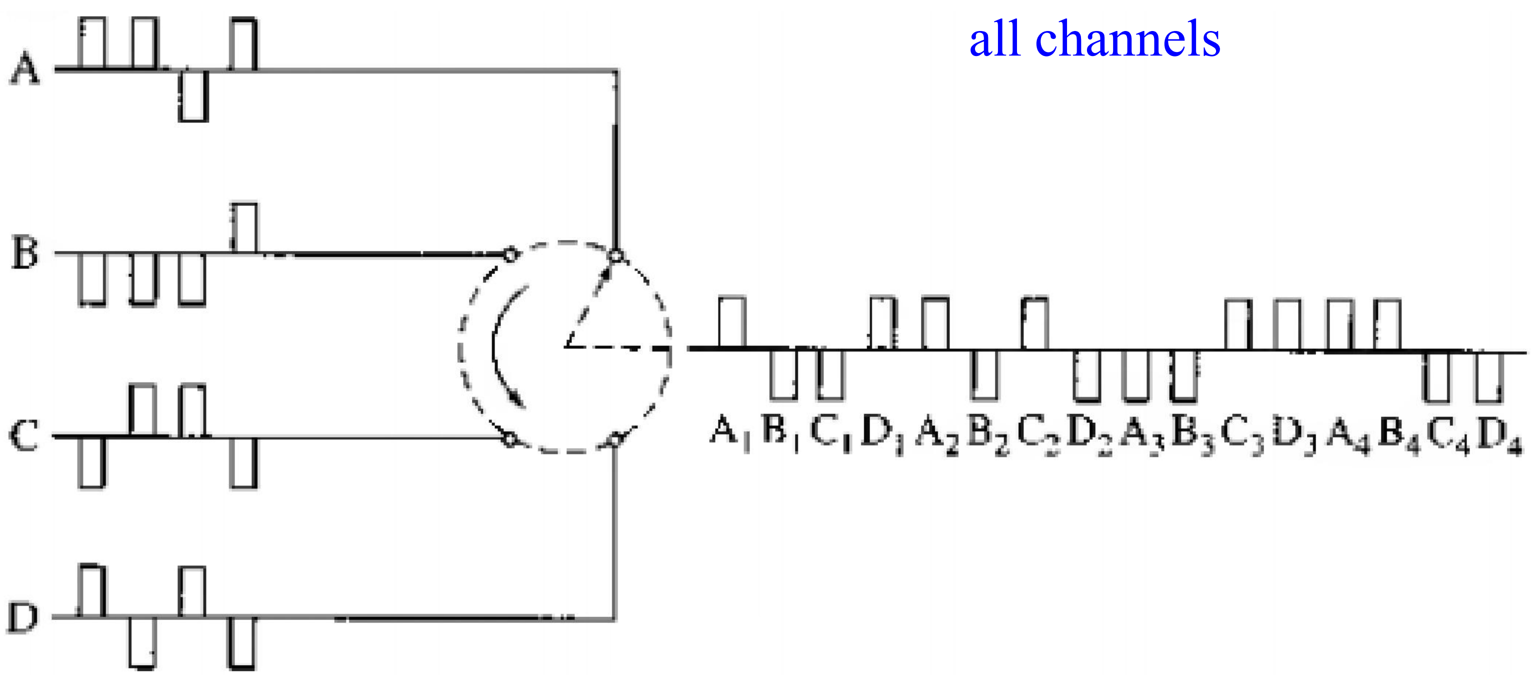


Identical bit rates for all channels

Used in: T1 system to generate DS1

Digital Multiplexing: Bit/Digit Interleaving

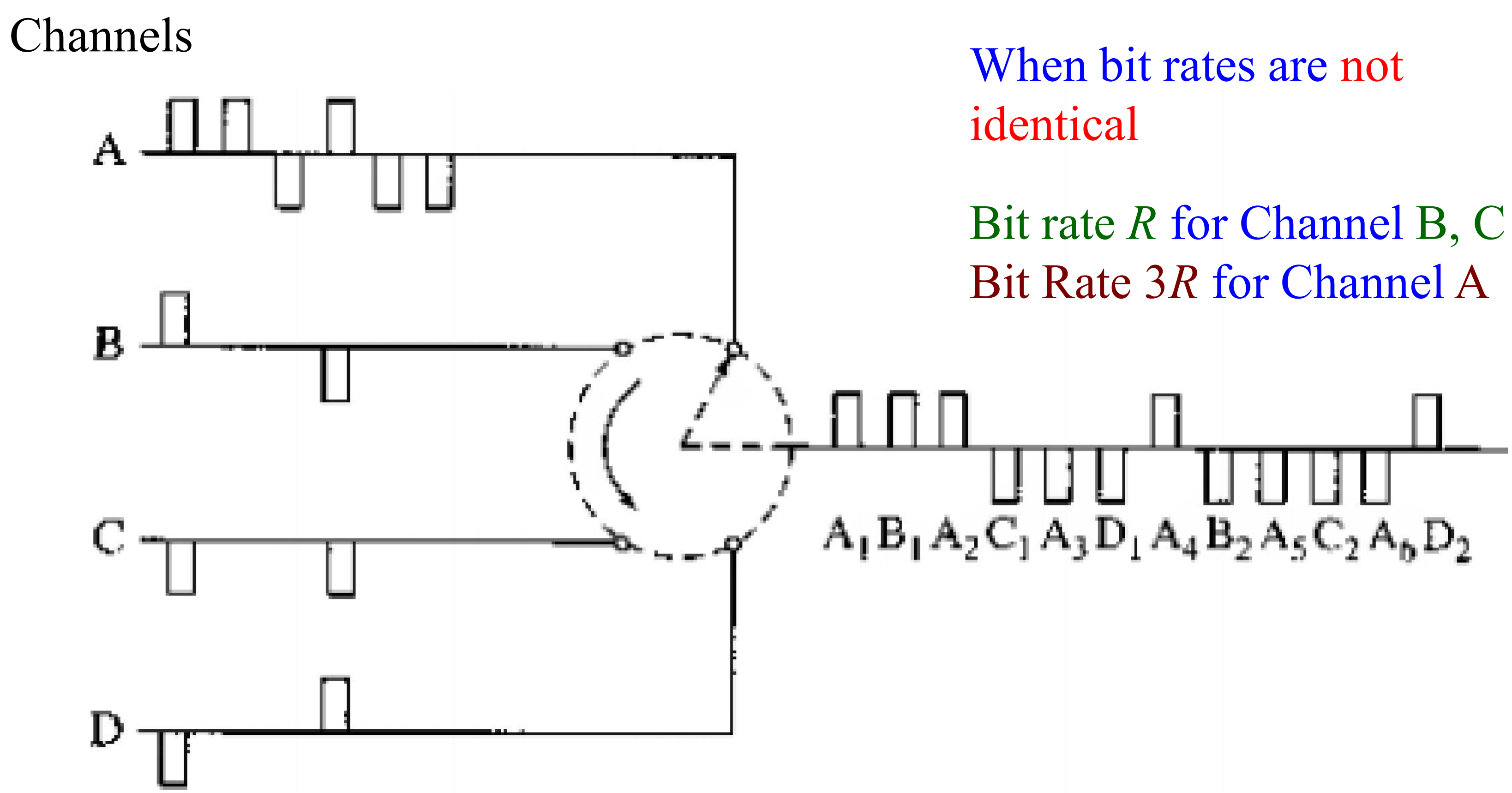
Channels



Identical bit rates for all channels

Used in: North American Digital Hierarchy

Digital Multiplexing: Bit/Digit Interleaving



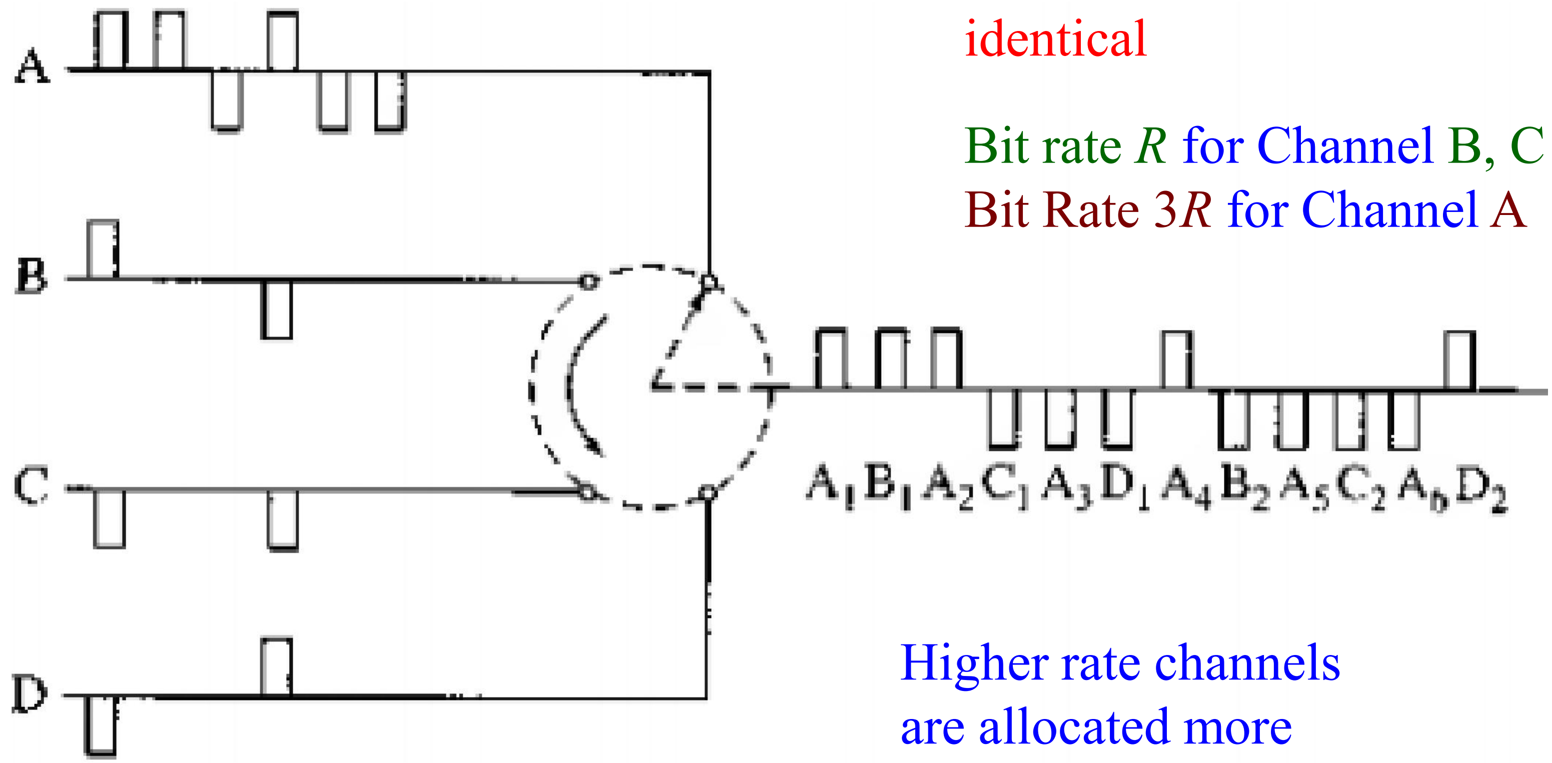
When bit rates are not identical

Bit rate R for Channel B, C and D

Bit Rate $3R$ for Channel A

Digital Multiplexing: Bit/Digit Interleaving

Channels



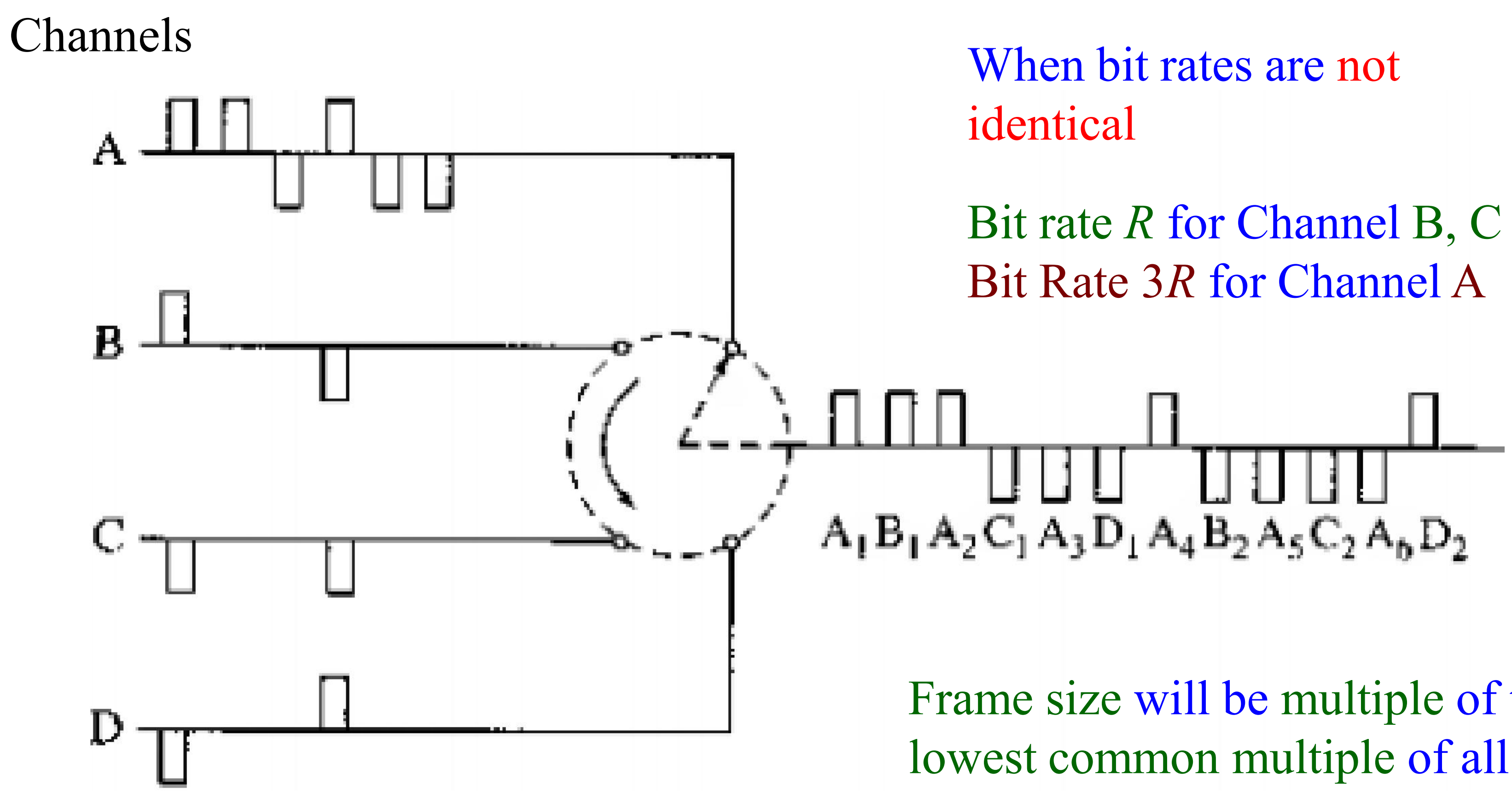
When bit rates are not identical

Bit rate *R* for Channel B, C and D

Bit Rate 3*R* for Channel A

Higher rate channels are allocated more slots

Digital Multiplexing: Bit/Digit Interleaving

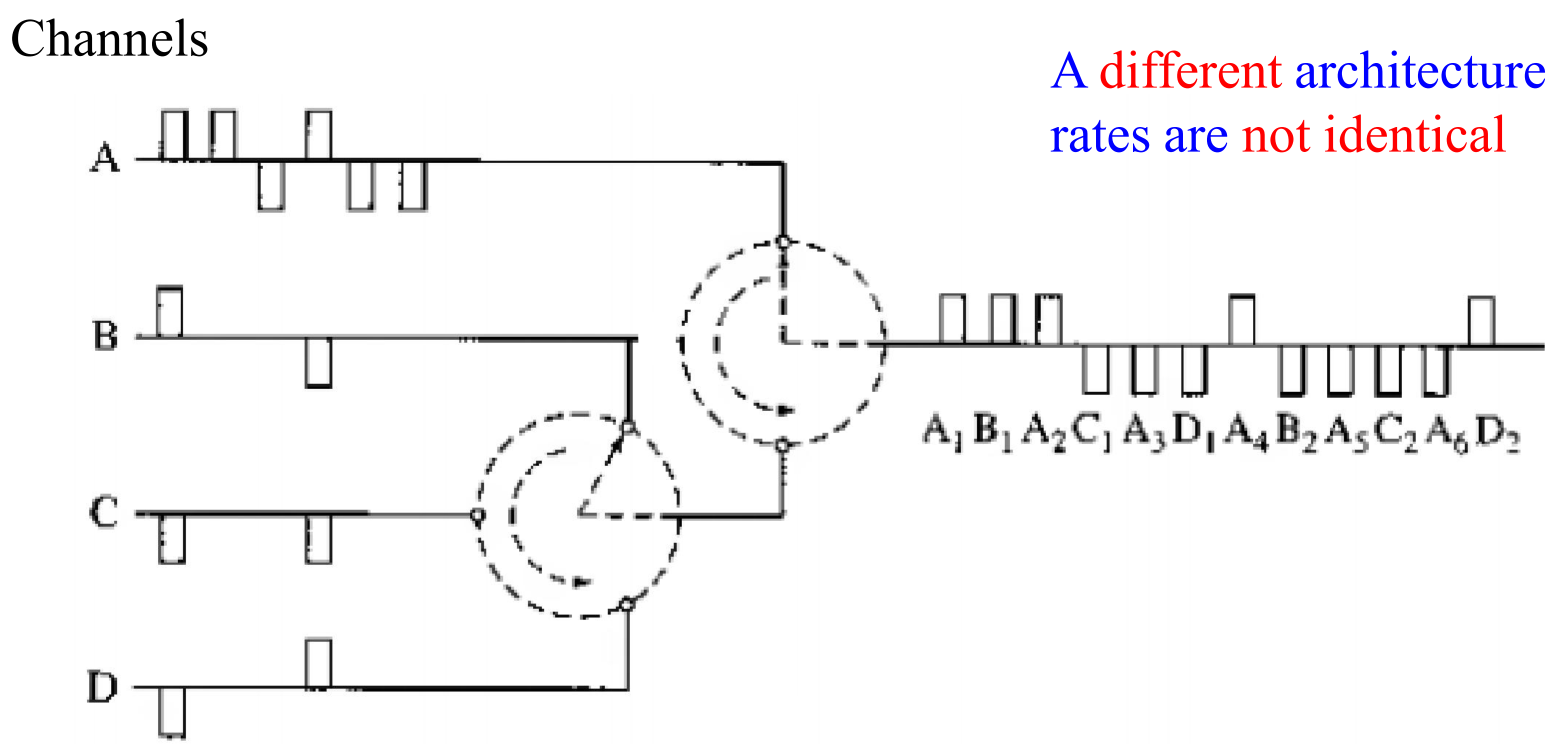


When bit rates are not identical

Bit rate R for Channel B, C and D
Bit Rate $3R$ for Channel A

Frame size will be multiple of the lowest common multiple of all rates

Digital Multiplexing: Bit/Digit Interleaving



A different architecture when bit rates are not identical

Digital Multiplexing: Receiver Side

- Data has to be divided and distributed in different channels

- Requires correct identification of bits which is correct identification of
  - beginning of each frame
  - slots in each frame
  - bit in a slot

Digital Multiplexing: Receiver Side

- Data has to be divided and distributed in different channels

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  - beginning of each frame
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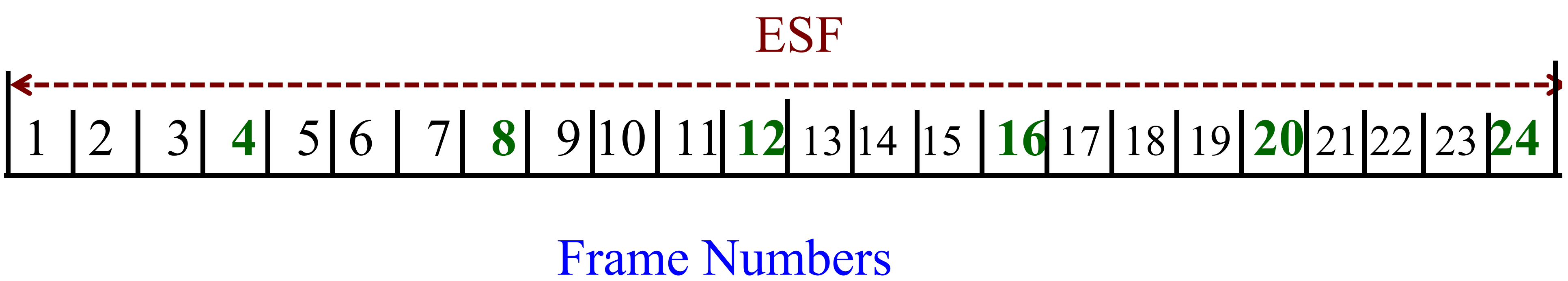
All these done by adding overhead bits which consist of

- Framing and synchronizing bits

Extended Superframe (ESF)

Structure of ESF



- Reduces framing bits: every 4th frame has a framing bit framing bits make a pattern **001011**: repeats in every 24 frames

DM1/2 Multiplexer Frame Format

$$M_0\ [48]\ C_A\ [48]\ F_0\ [48]\ C_A\ [48]\ C_A\ [48]\ F_1\ [48]$$

$$M_1\ [48]\ C_B\ [48]\ F_0\ [48]\ C_B\ [48]\ C_B\ [48]\ F_1\ [48]$$

$$M_1\ [48]\ C_C\ [48]\ F_0\ [48]\ C_C\ [48]\ C_C\ [48]\ F_1\ [48]$$

$$M_1\ [48]\ C_D\ [48]\ F_0\ [48]\ C_D\ [48]\ C_D\ [48]\ F_1\ [48]$$

- DM1/2: converts 4 × DS1 (1.544 Mbps) to 1 × DS2 (6.312 Mbps)
- Bit-by-bit interleaving of 4 channels of 1.544 Kbps
- The frame has 4 sub-frames
- Each sub-frame has 6 overhead bits

DM1/2 Multiplexer Frame Format

M_0	[48]	C_A	[48]	F_0	[48]	C_A	[48]	C_A	[48]	F_1	[48]
M_1	[48]	C_B	[48]	F_0	[48]	C_B	[48]	C_B	[48]	F_1	[48]
M_1	[48]	C_C	[48]	F_0	[48]	C_C	[48]	C_C	[48]	F_1	[48]
M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

All [48]'s : Data bits, interleaved from 4 channels

C_A, C_B, C_C, C_D, : Stuffing bits

F_0, F_1 : Main framing bits

M_0, M_1 : Sub-framing bits

DM1/2 Multiplexer Frame Format

M_0 [48] C_A [48] F_0 [48] C_A [48] C_A [48] F_1 [48]

M_1 [48] C_B [48] F_0 [48] C_B [48] C_B [48] F_1 [48]

M_1 [48] C_C [48] F_0 [48] C_C [48] C_C [48] F_1 [48]

M_1 [48] C_D [48] F_0 [48] C_D [48] C_D [48] F_1 [48]

Bit wise interleaved from 4 channels

DM1/2 Multiplexer Frame Format

M_0	[48]	C_A	[48]	F_0	[48]	C_A	[48]	C_A	[48]	F_1	[48]
M_1	[48]	C_B	[48]	F_0	[48]	C_B	[48]	C_B	[48]	F_1	[48]
M_1	[48]	C_C	[48]	F_0	[48]	C_C	[48]	C_C	[48]	F_1	[48]
M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

Total Data bits = $48\times6\times4$ = 1152 bits

Total overhead bits = 6×4 = 24 bits

Efficiency = 1152/1176 $\approx$ 98%

DM1/2 Multiplexer Frame Format

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M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

M_0, = F_0 = 0

M_1, = F_1 = 1

Values of C_A, C_B, C_C, C_D, depend on stuffing types

DM1/2 Multiplexer Frame Format

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M_1 [48] C_D [48] F_0 [48] C_D [48] C_D [48] F_1 [48]

Framing bits make a periodic pattern 010101… : used to synchronize on the frame

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$M_0M_1M_1M_1 = 0111$ is used to identify sub-frames.

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$M_0M_1M_1M_1 = 0111$ is used to identify sub-frames.

Pattern $M_0M_1M_1M_1 = 0111$ also confirms genuine $F_0F_1F_0F_1$ pattern

Complication in Switching & Signaling

- All channels are NOT active always: some transmits, some are idle
- multiplexer’s slots may be underutilized
- Additional input channels are allowed

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Complexity

- NO guaranty to hold: No. of active channels ≤ No. of available slots
- Complicated switching operations

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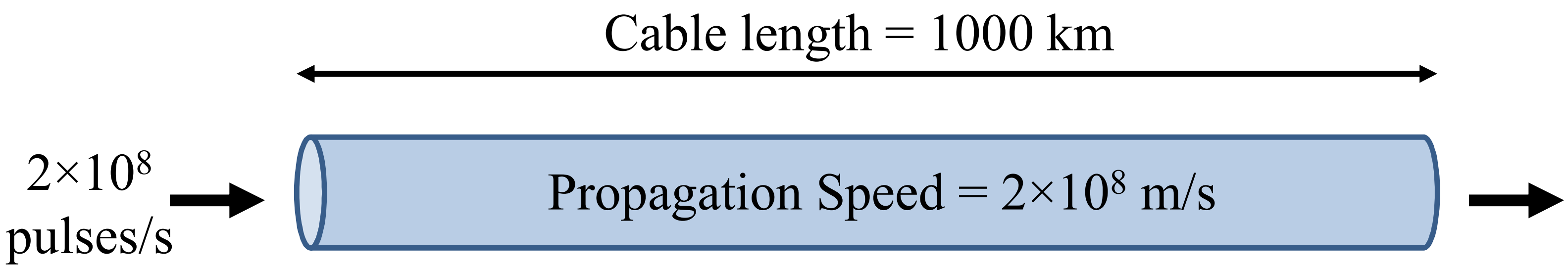
Management

- Use previous statistics to minimize the overload

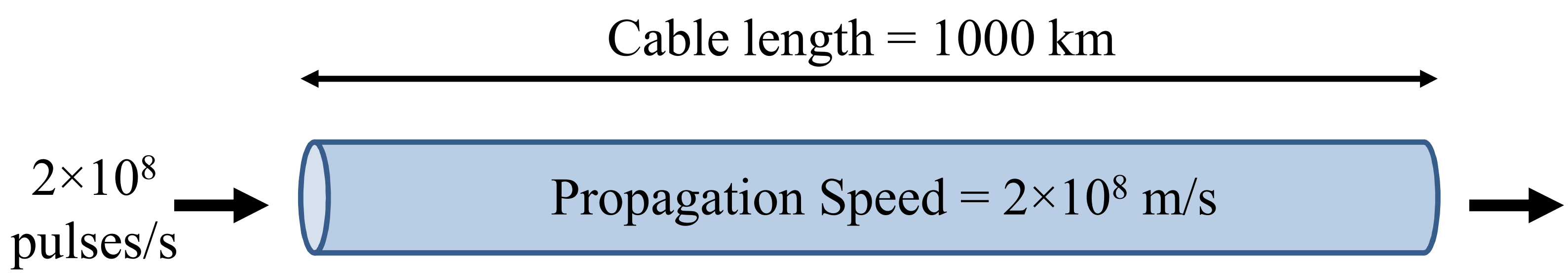
Asynchronous Channels & Bit Stuffing

- Synchronization is lost due to different reasons
  - variation in oscillation of oscillators
  - variation in temperature of carrying cables

Asynchronous Channels & Bit Stuffing



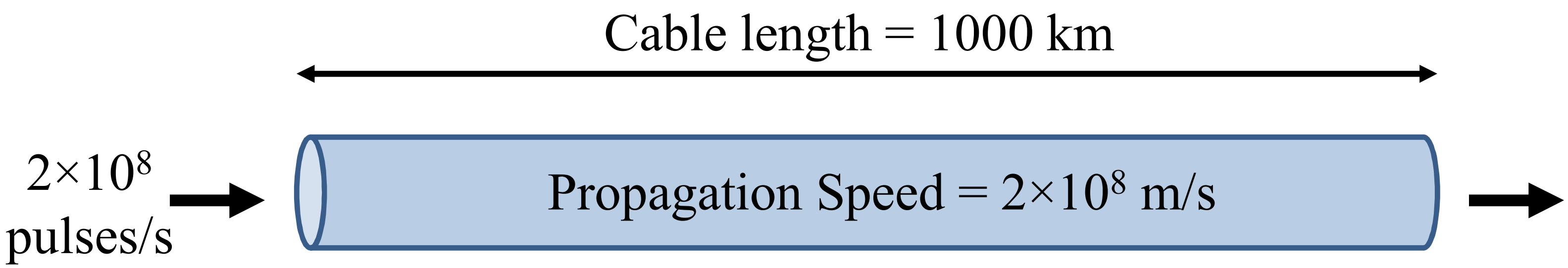
Asynchronous Channels & Bit Stuffing



Propagation time for a pulse through the cable = 1000 km / (2×10^8 m/s)

=1/200 second

Asynchronous Channels & Bit Stuffing



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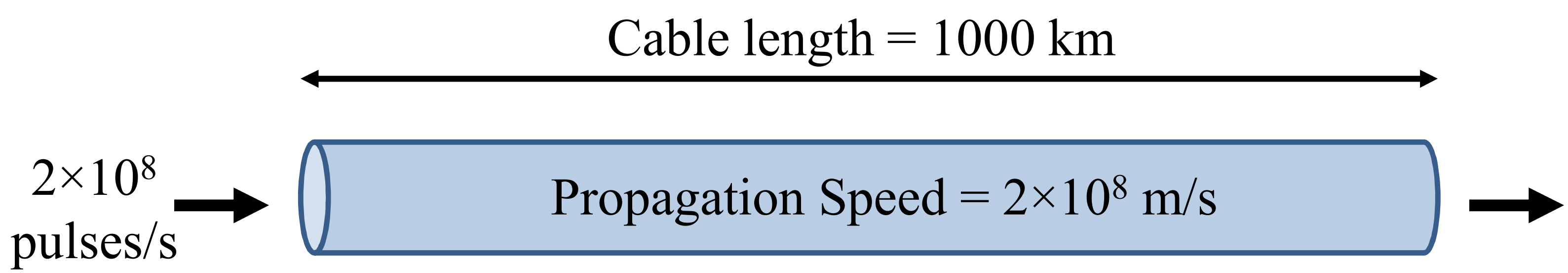

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1^0 F increase in cable temperature => 0.01% increase in propagation speed

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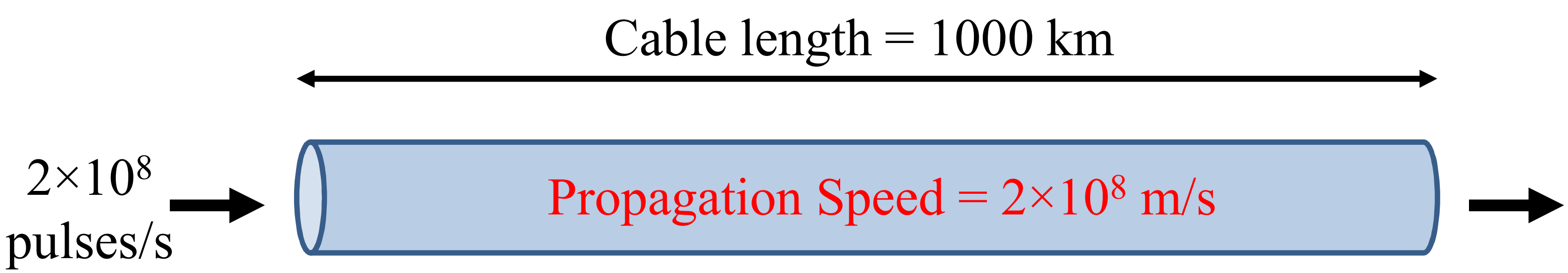

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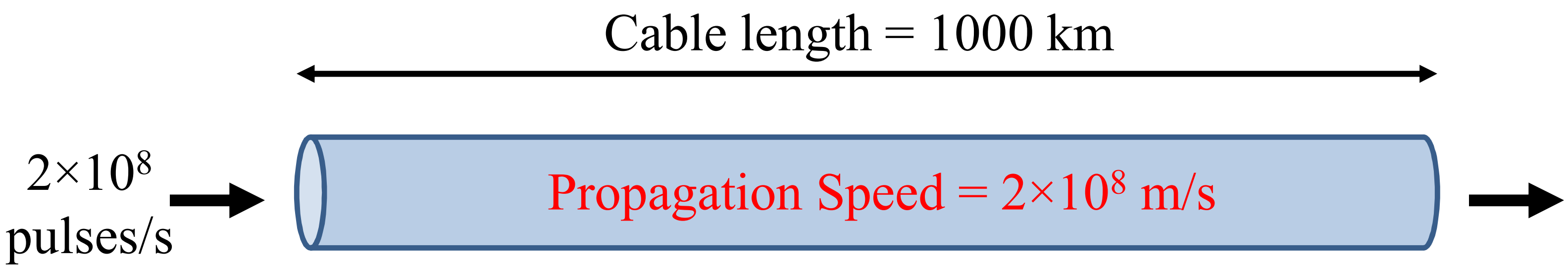
++

1^0 F increase in cable temperature => 0.01% increase in propagation speed

Asynchronous Channels & Bit Stuffing



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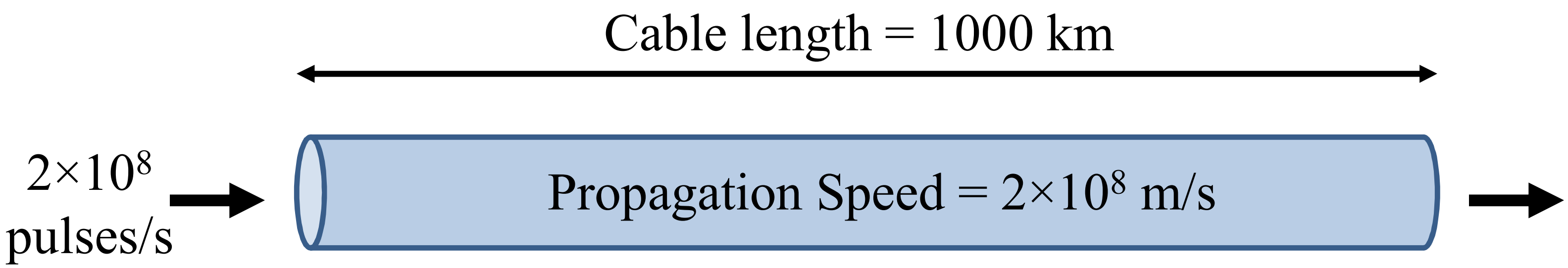
++

1^0 F increase in cable temperature => 0.01% increase in propagation speed

Extra pulses must be stored somewhere

Asynchronous Channels & Bit Stuffing



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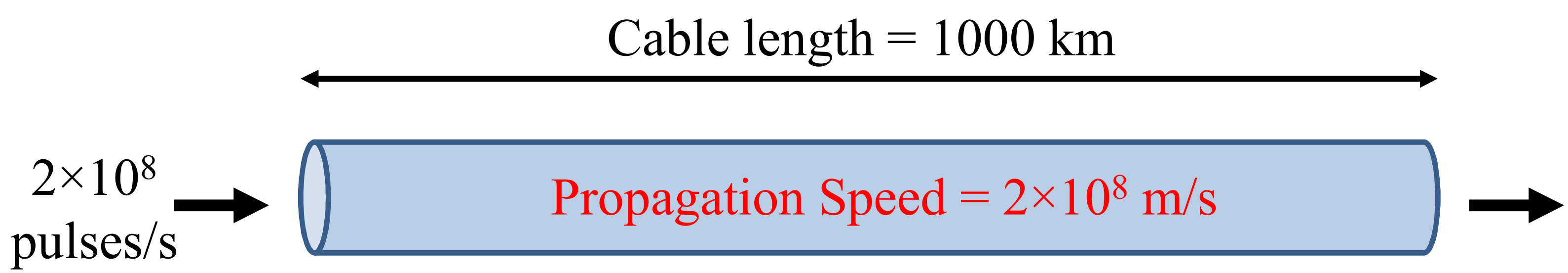
=1/200 second

No. of Pulses in transit within this time = 1/200 second / (2×10^8 pulses/s)
=10^6 pulses

drop in cable temperature => rate of received pulses drops => vacant slots in multiplexer => vacant slots to be stuffed with dummy bits

Asynchronous Channels & Bit Stuffing



Findings:

- Even synchronous multiplexer system may receive bits asynchronously
- We need
  - storage for / removal of extra bits
  - Stuffing for vacant slots

Asynchronous Channels & Bit Stuffing

Stuffing classification

- Positive
- Negative
- Positive/negative

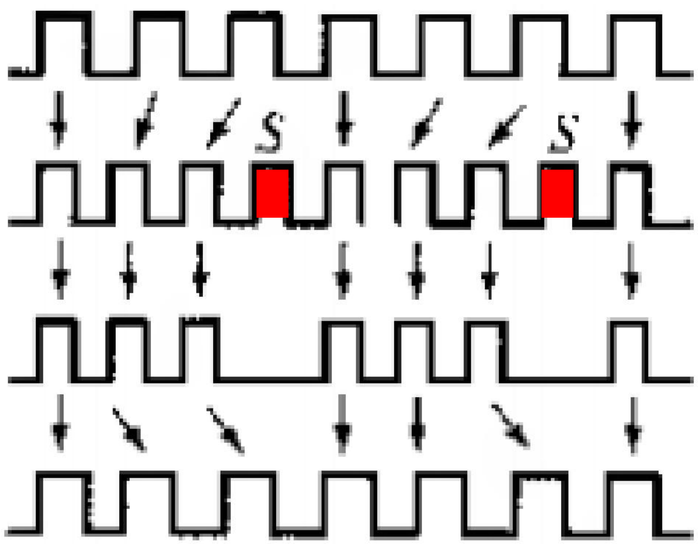
Asynchronous Channels & Bit Stuffing

Positive Stuffing

- Multiplexer's rate > total max rates of all incoming channels
- Time slots in multiplexer remain vacant at some stage
- Pulse stuffing becomes necessary
- Stuffed bit and position are encoded using C-bits

Asynchronous Channels & Bit Stuffing

Positive Stuffing



Input signal to multiplexer

Transmitted signal with stuffing

Un-stuffed signal

Output signal with smoothing

Asynchronous Channels & Bit Stuffing

Negative Stuffing

- Multiplexer's rate < total rates of all incoming channels
- Time slots in multiplexer becomes overloaded at some stage
- Multiplexer cannot accommodate all incoming signals
- Information of left-out pulses and position is transmitted through overhead bits

Asynchronous Channels & Bit Stuffing

Positive/Negative Stuffing

- Combination of positive and negative stuffing
- Nominal rate of multiplexer = total nominal rate of incoming channels
- positive stuffing at sometimes and negative at others
- All information is transmitted through overhead bits

Transmitting Stuffing Information through C-bits

M_0 [48] C_A [48] F_0 [48] C_A [48] C_A [48] F_1 [48]

M_1 [48] C_B [48] F_0 [48] C_B [48] C_B [48] F_1 [48]

M_1 [48] C_C [48] F_0 [48] C_C [48] C_C [48] F_1 [48]

M_1 [48] C_D [48] F_0 [48] C_D [48] C_D [48] F_1 [48]

- Four incoming channels: A, B, C, D
- Only 1 bit stuffing is allowed per channel per frame

Transmitting Stuffing Information through C-bits

M_0	[48]	C_A	[48]	F_0	[48]	C_A	[48]	C_A	[48]	F_1	[48]
M_1	[48]	C_B	[48]	F_0	[48]	C_B	[48]	C_B	[48]	F_1	[48]
M_1	[48]	C_C	[48]	F_0	[48]	C_C	[48]	C_C	[48]	F_1	[48]
M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

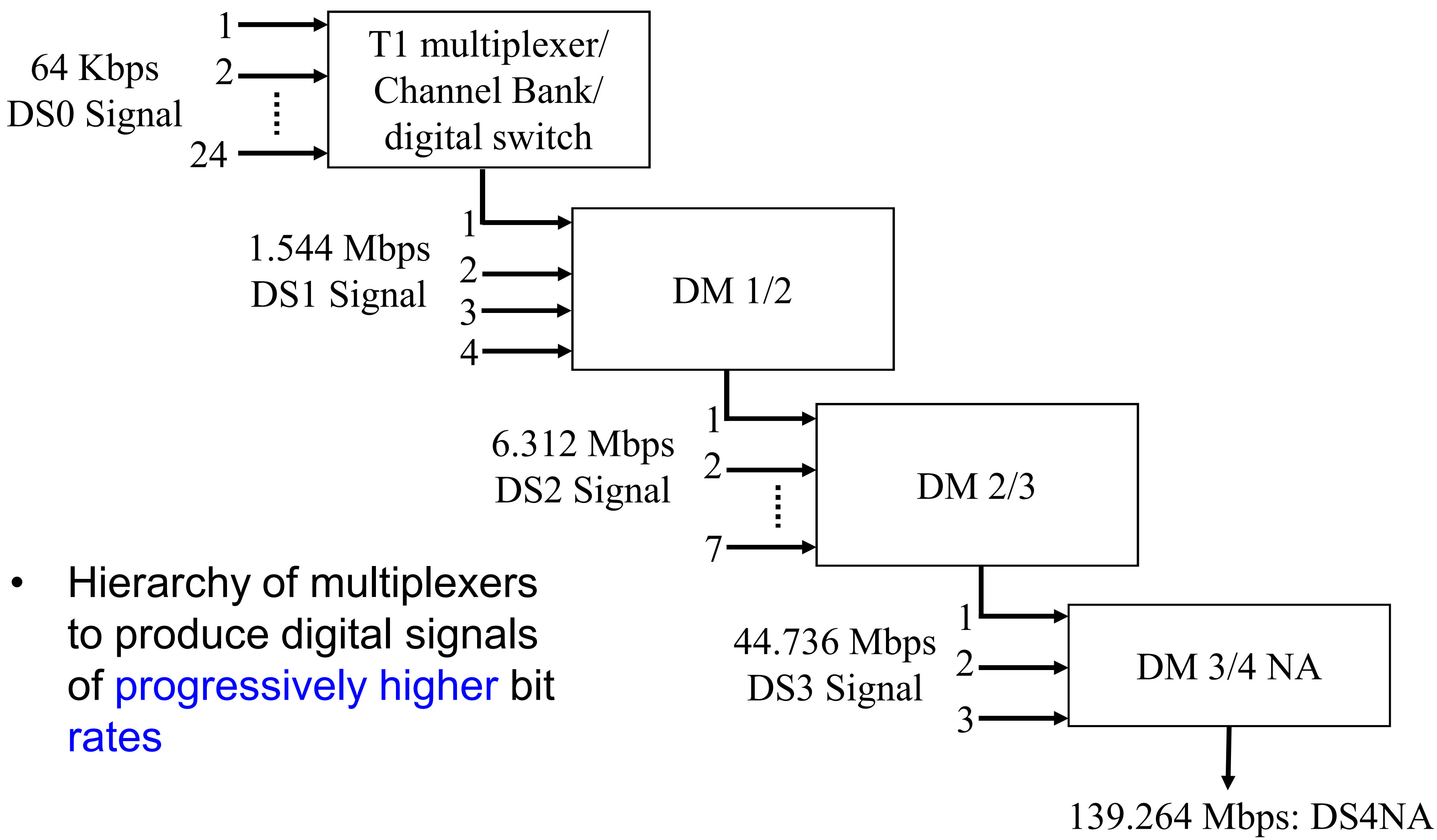
Channel	Stuffing	No-Stuffing
A	$C_A C_A C_A = 111$	$C_A C_A C_A = 000$
B	$C_B C_B C_B = 111$	$C_B C_B C_B = 000$
C	$C_C C_C C_C = 111$	$C_C C_C C_C = 000$
D	$C_D C_D C_D = 111$	$C_D C_D C_D = 000$

Transmitting Stuffing Information through C-bits

M_0	[48]	C_A	[48]	F_0	[48]	C_A	[48]	C_A	[48]	F_1	[48]
M_1	[48]	C_B	[48]	F_0	[48]	C_B	[48]	C_B	[48]	F_1	[48]
M_1	[48]	C_C	[48]	F_0	[48]	C_C	[48]	C_C	[48]	F_1	[48]
M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

- The stuffed bit is the first bit following F_1 in the corresponding sub-frame

Plesiochronous (almost Synchronous) Digital Hierarchy



- Hierarchy of multiplexers to produce digital signals of progressively higher bit rates

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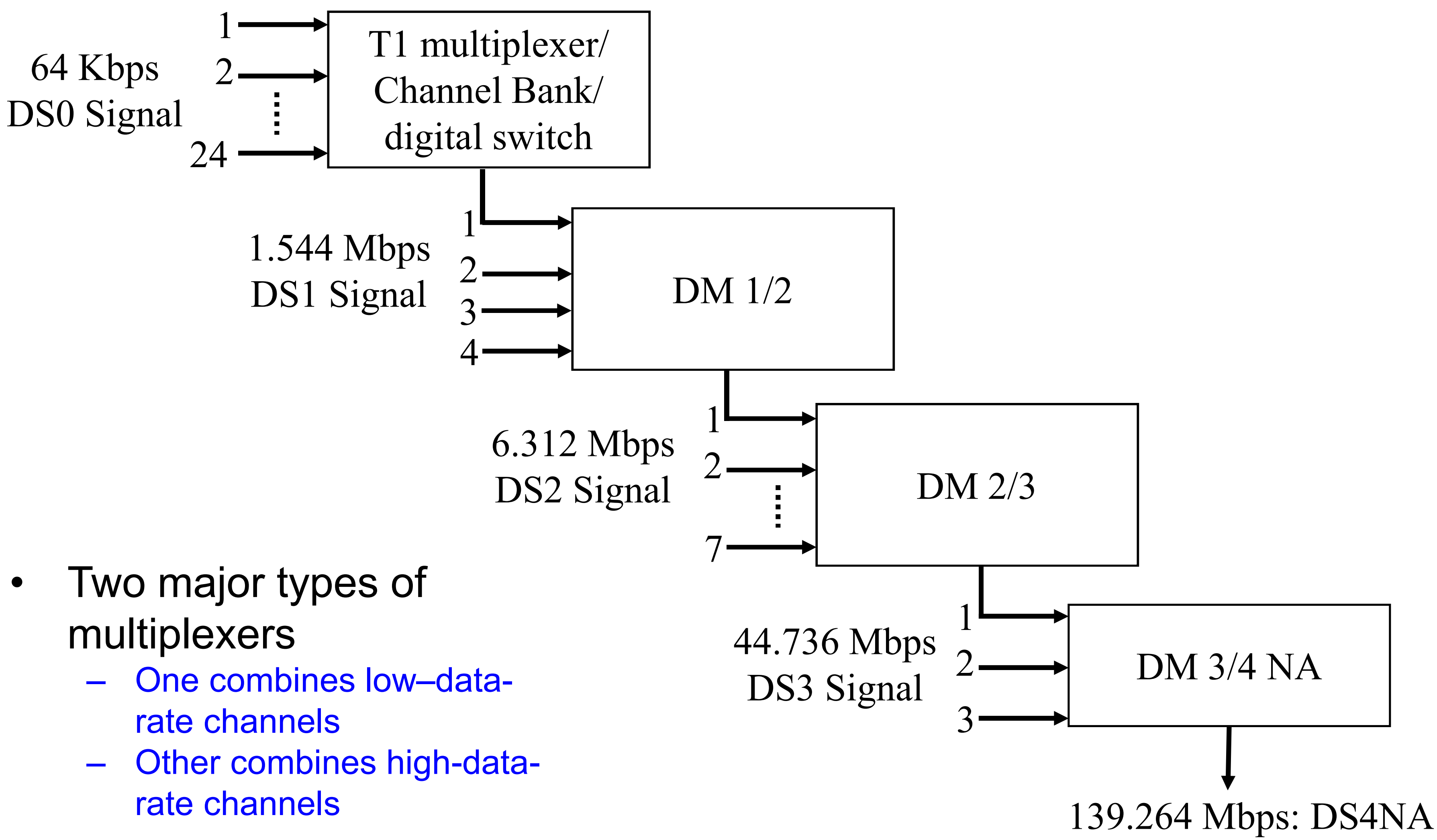
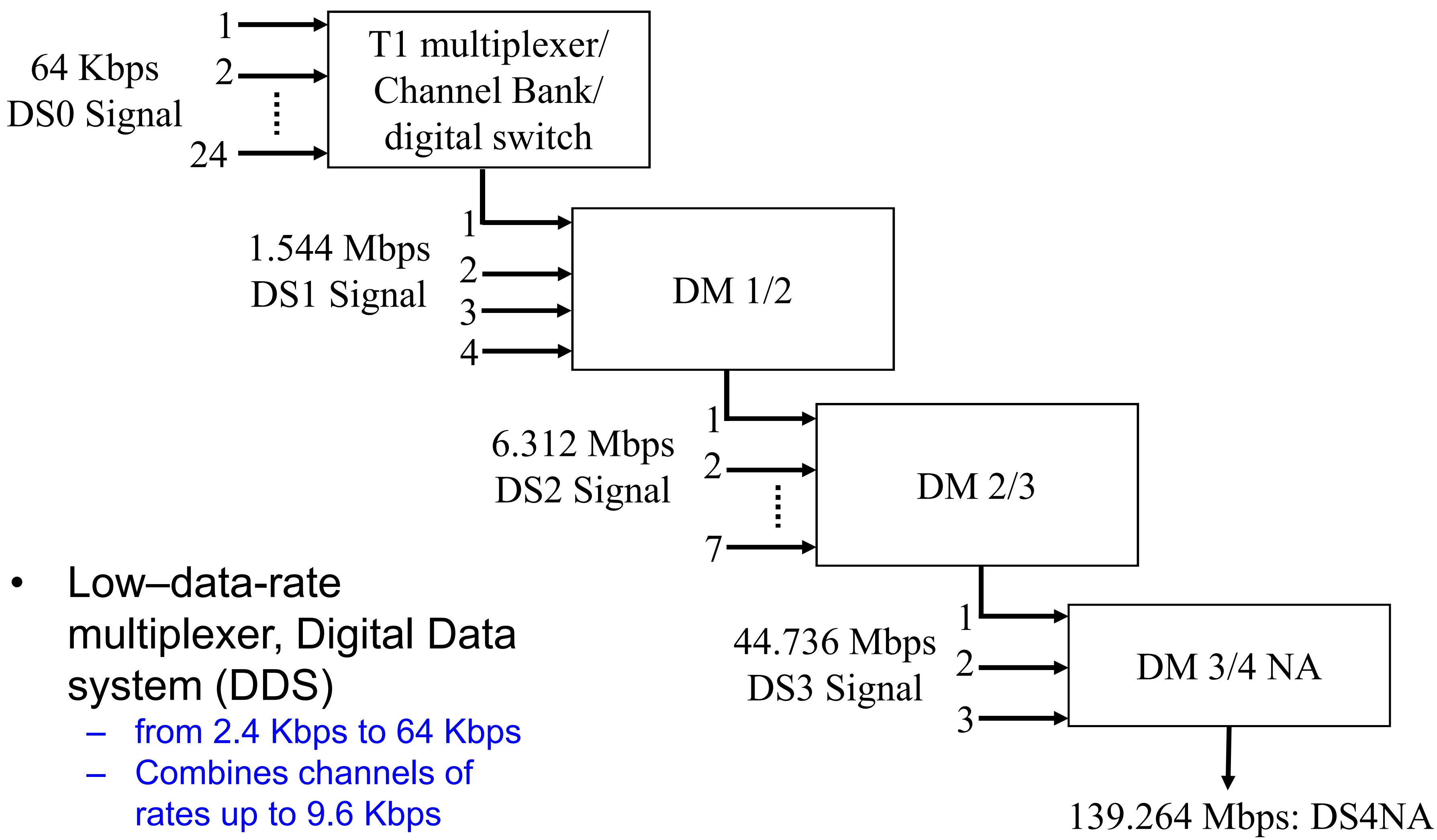


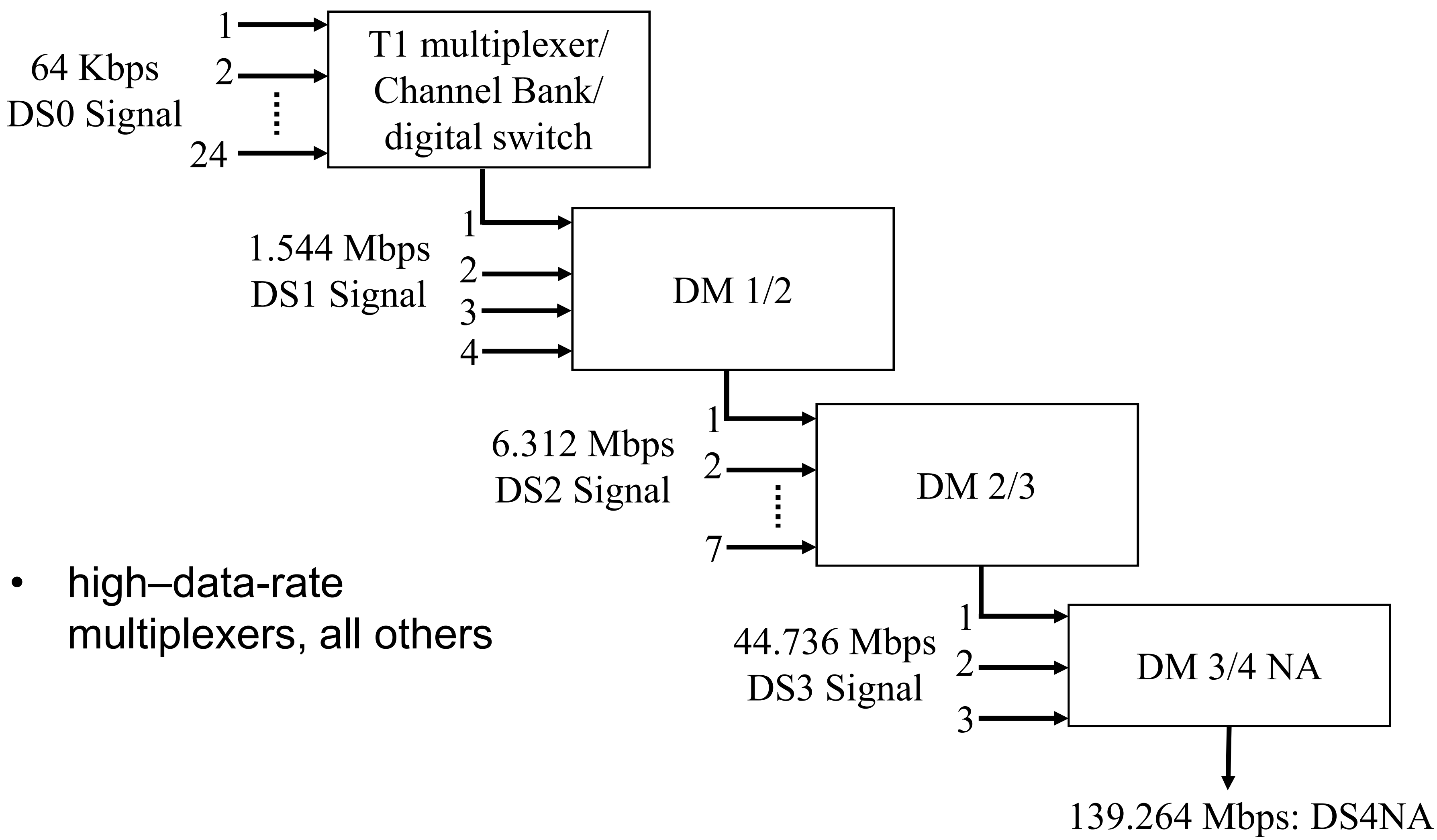
- Two major types of multiplexers
  - One combines low–data-rate channels
  - Other combines high-data-rate channels

Plesiochronous (almost Synchronous) Digital Hierarchy



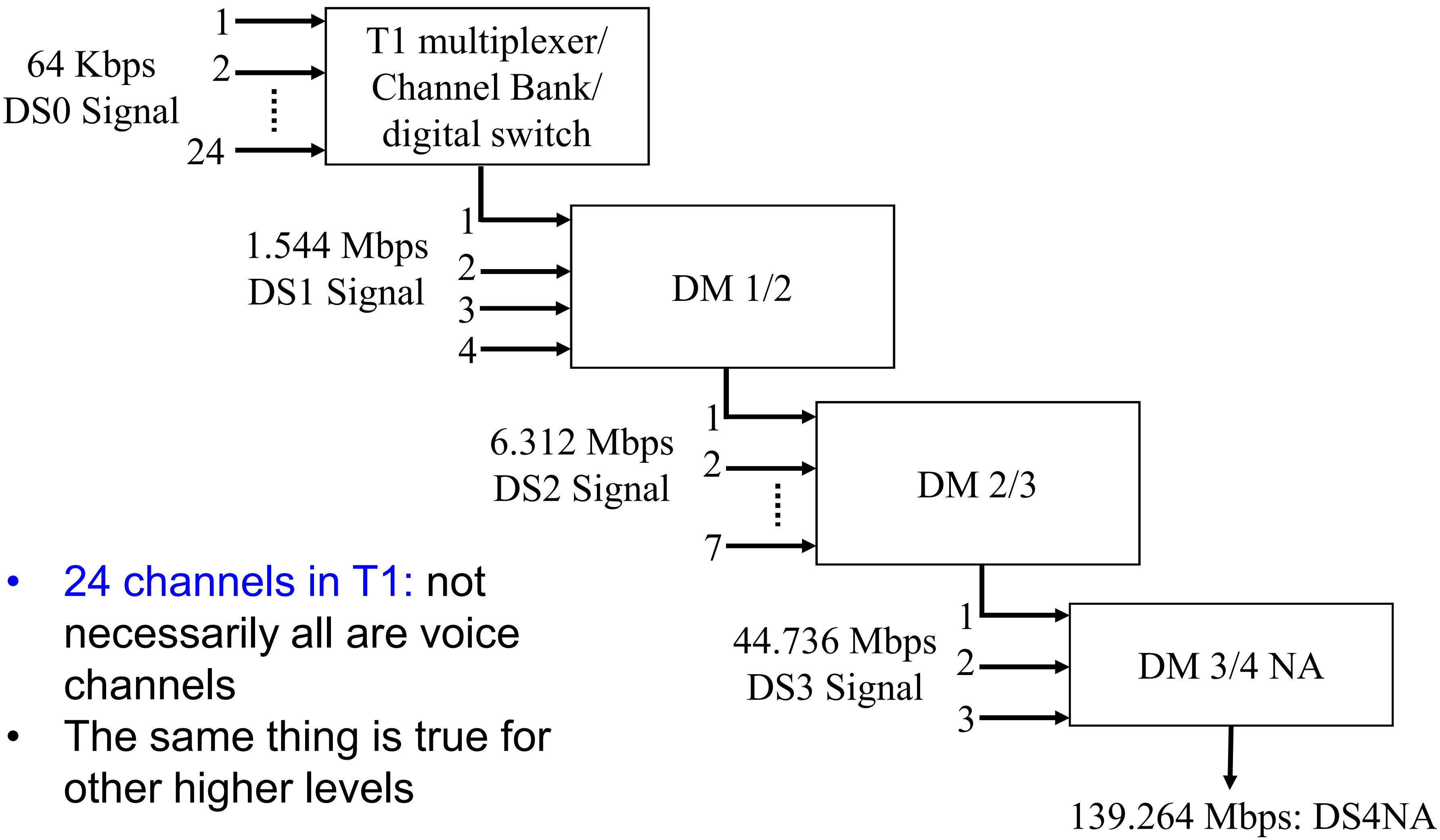
- Low–data-rate multiplexer, Digital Data system (DDS)
  - from 2.4 Kbps to 64 Kbps
  - Combines channels of rates up to 9.6 Kbps

Plesiochronous (almost Synchronous) Digital Hierarchy



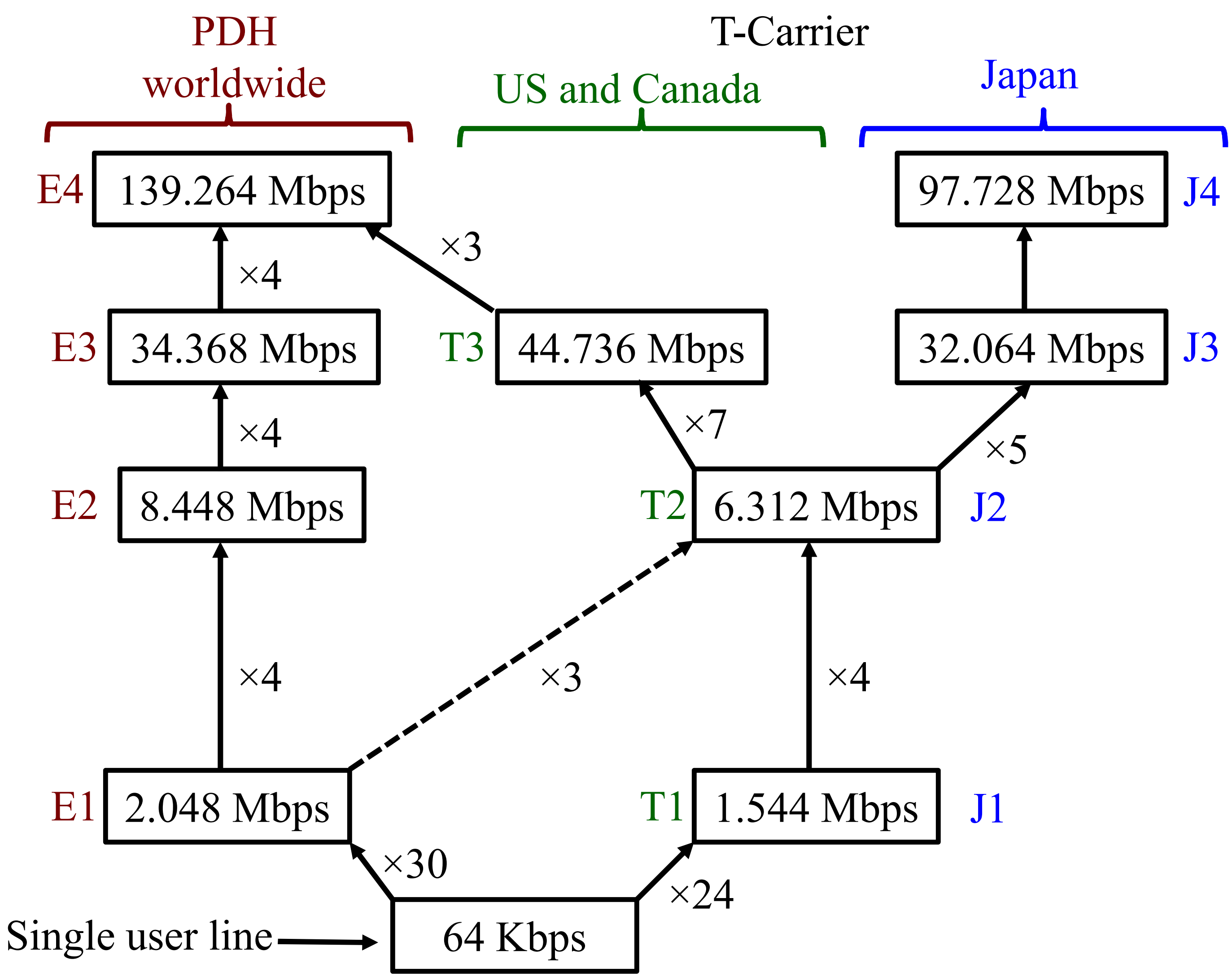
- high–data-rate multiplexers, all others

Plesiochronous (almost Synchronous) Digital Hierarchy



- 24 channels in T1: not necessarily all are voice channels
- The same thing is true for other higher levels

Other Plesiochronous Digital Hierarchies



Differential Pulse Code Modulation (DPCM)

PCM and Transmission Channel Bandwidth: Example-1

Analog signal bandwidth => digital data rate => Channel Bandwidth requirement

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility)

Signal bandwidth, $B = 3400$ Hz
Sampling frequency, $f_s = 8000 > 2B$

PCM Quantization level, $L = 256$ (8 bits)

Data rate = 8000*8 =64000 pulse/second = 64 Kbps

PCM and Transmission Channel Bandwidth: Example-2

Example 2: data rate for compact disc

High Fidelity is required!

Audio signal b/w = 20 KHz

$B = 20000$ Hz
$f_s = 441000 \text{ Hz} > 2B$

Quantization level, $L = 65{,}536$ (16 bits)

Data rate = 44100*16 = 1.4 **Mbps =1400 Kbps**

PCM and Transmission Channel Bandwidth

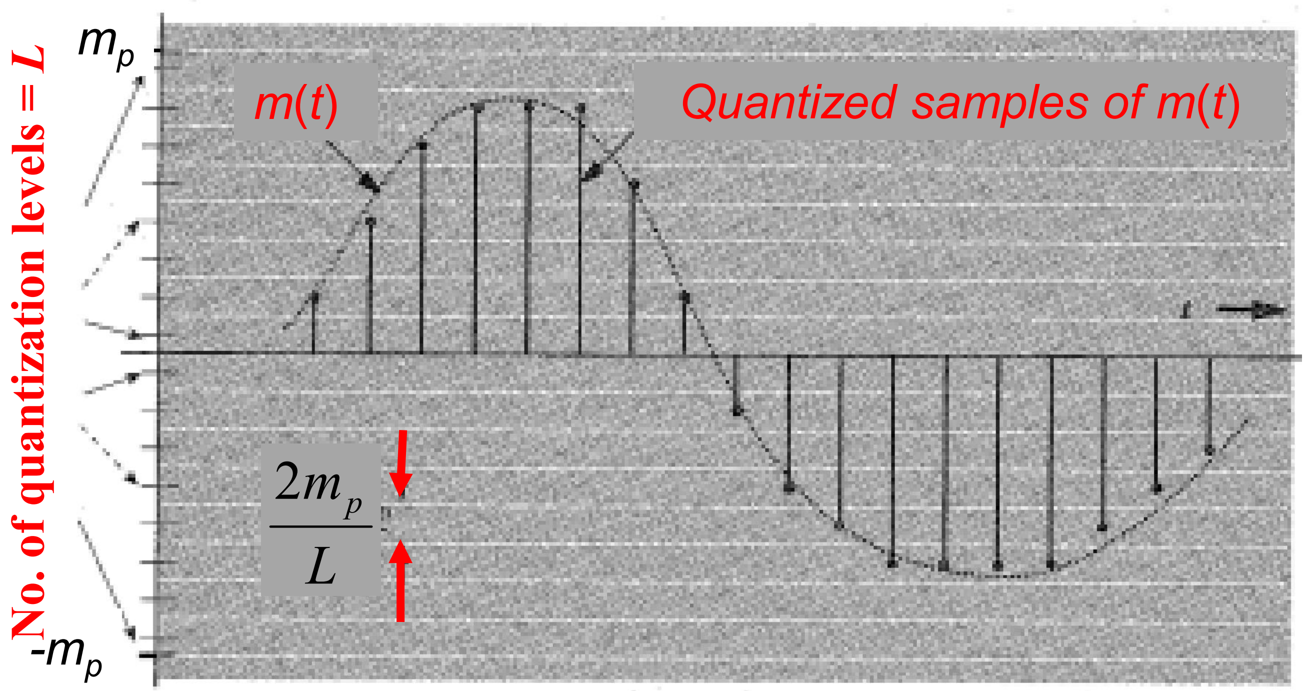
Findings

- PCM generates too many bits
- Requires high bandwidth to transmit them

PCM and Transmission Channel Bandwidth

Causes:

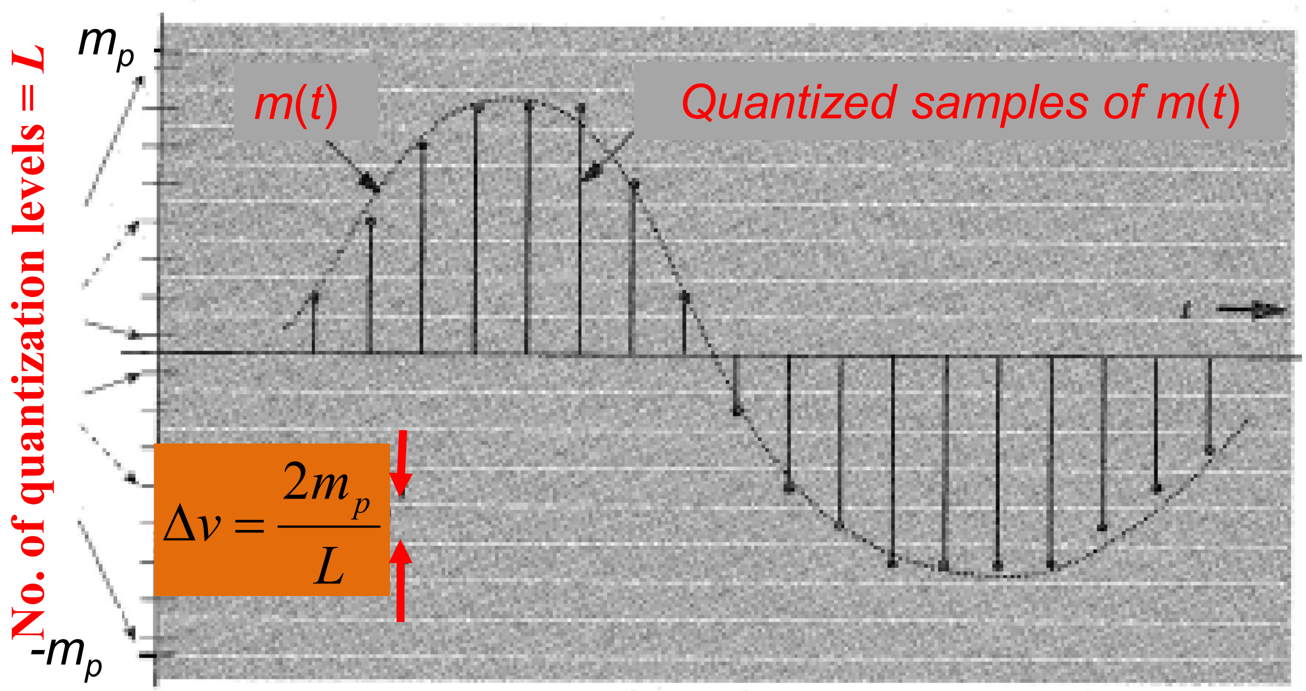
Every sample is represented by $n = \log_2 L$ bits



PCM and Transmission Channel Bandwidth

Causes:

Every sample is represented by $n = \log_2 L$ bits



decreasing L

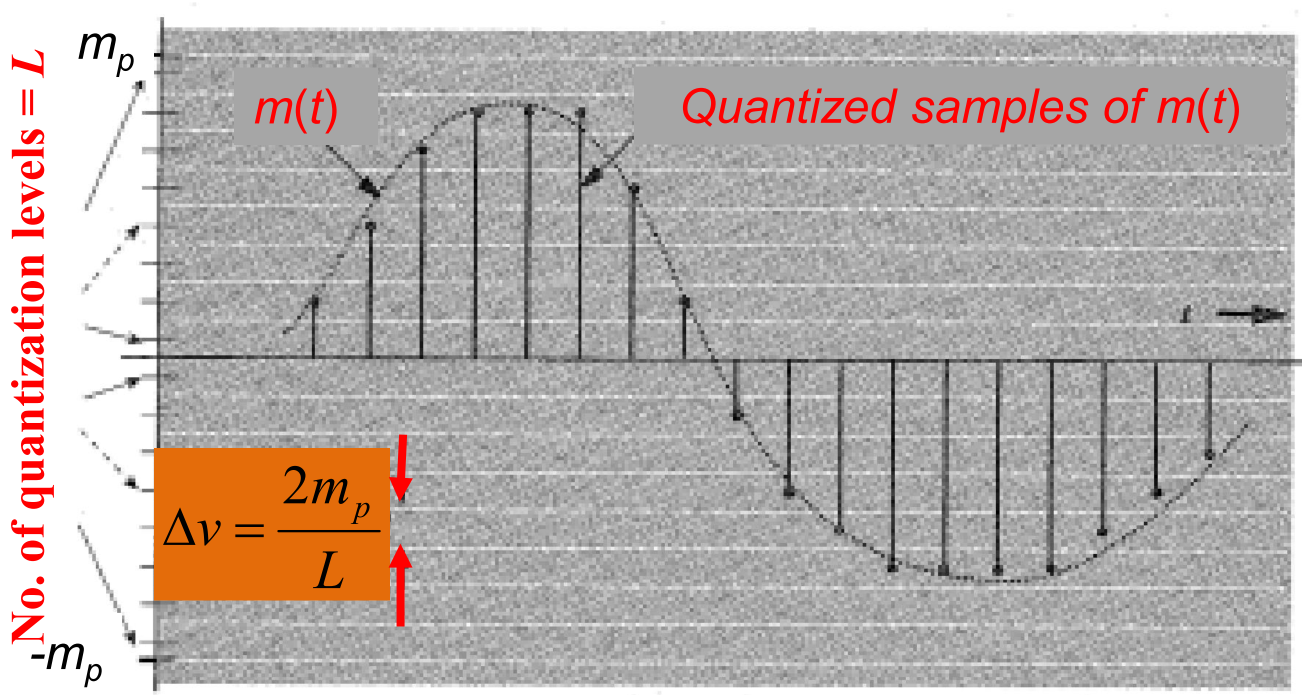
increasing Δv

increasing NOISE

PCM and Transmission Channel Bandwidth

Causes:

Every sample is represented by $n = \log_2 L$ bits



decreasing L

increasing Δv

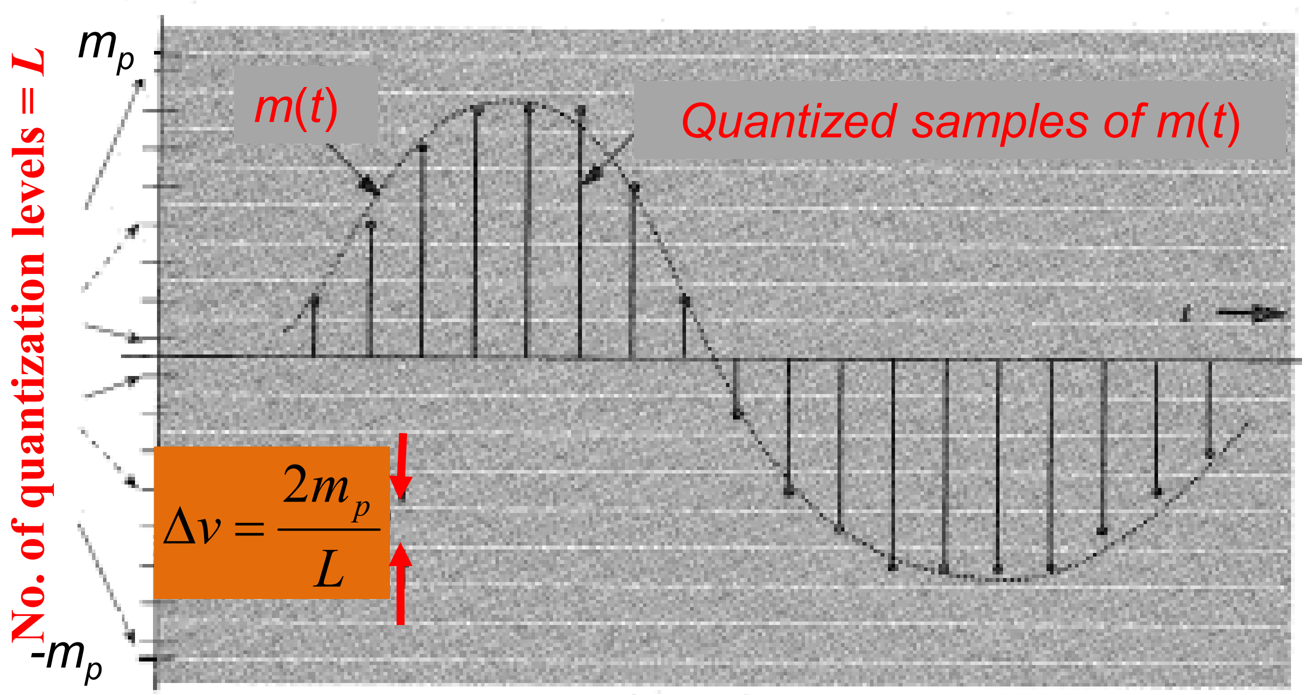
increasing NOISE

Because quantization noise $N_0 = (\Delta v)^2/12$

PCM and Transmission Channel Bandwidth

WAY OUT

- Successive samples are NOT independent, can be predicted
- Redundancy (CORRELATION) exists among neighboring samples
- Difference can be transmitted instead of actual sample



PCM and Transmission Channel Bandwidth

WAY OUT

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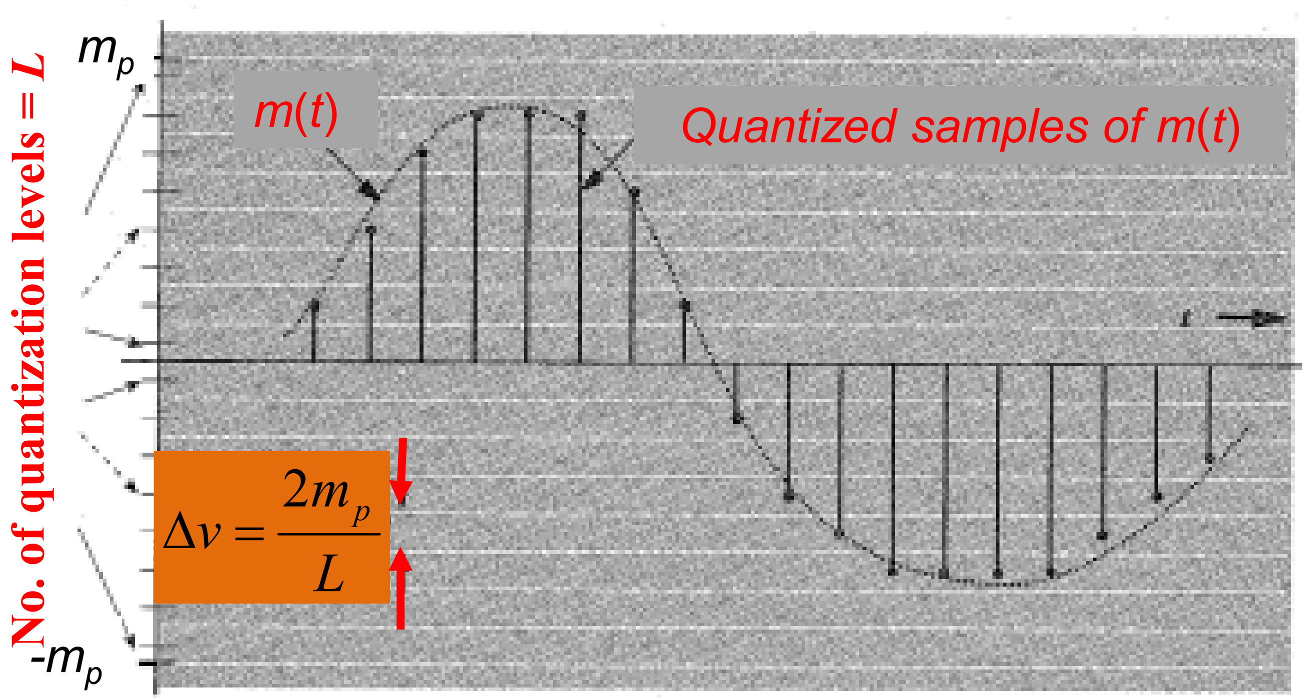

$d[k] = m[k] - m[k\text{-}1]$

Due t o redundancy,

$d[k] <<< m[k]$

PCM and Transmission Channel Bandwidth

WAY OUT

- Successive samples are NOT independent, can be predicted
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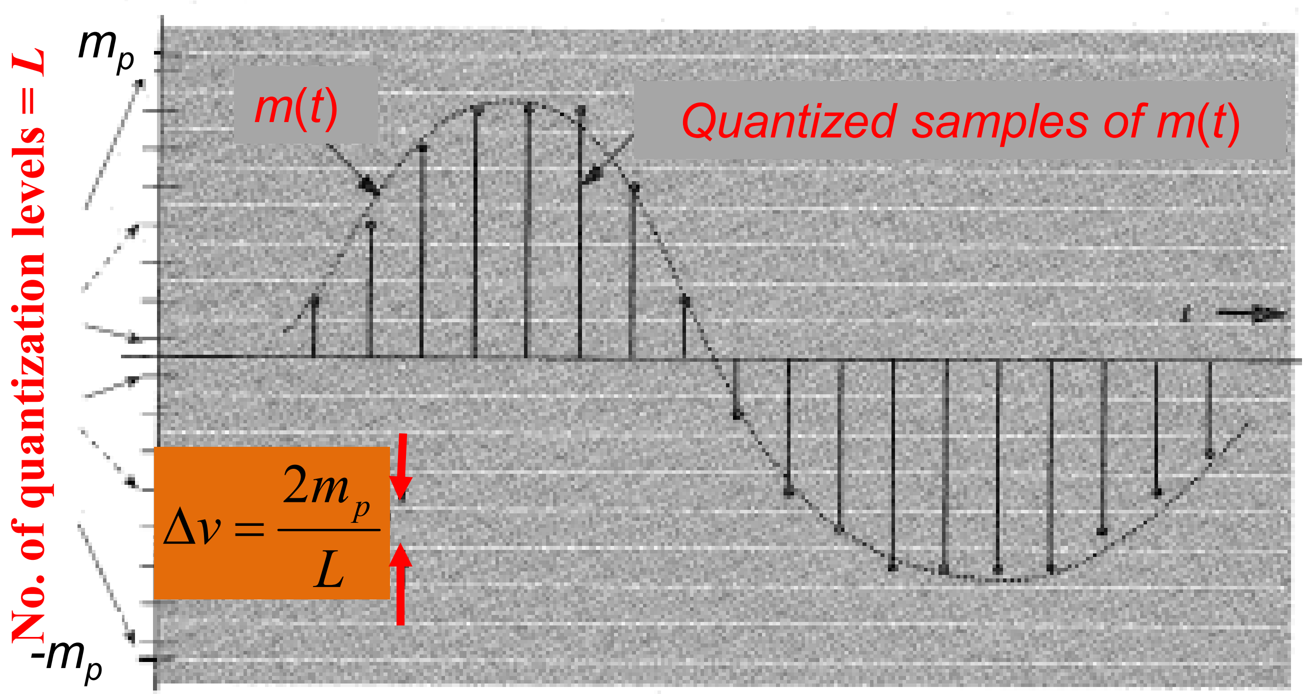


$d[k] = m[k] - m[k\text{-}1]$

decreases peak d_p

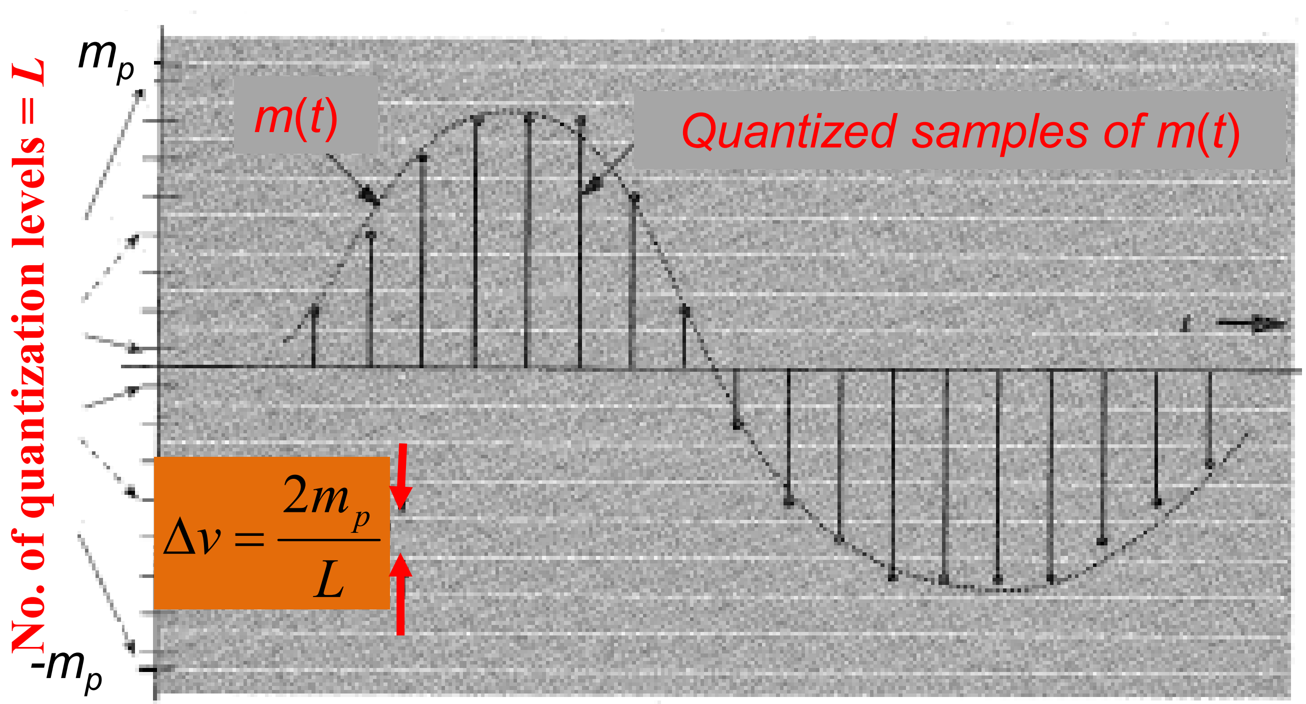
decreases Δv_d

decreases NOISE

PCM and Transmission Channel Bandwidth

Alternatively,

For same NOISE or SNR level, samples can be transmitted by fewer $n = \log_2 L$ bits



$d[k] = m[k] - m[k-1]$

decreases peak d_p

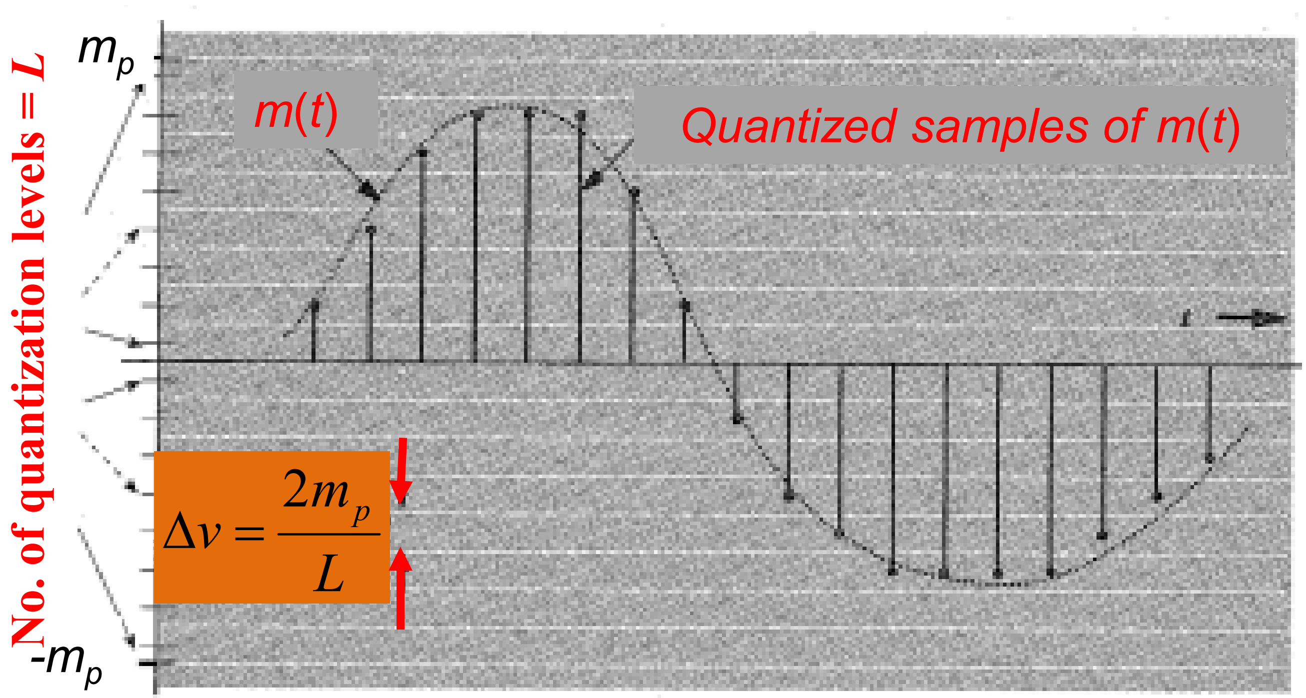
decreases Δv_d

decreases NOISE

PCM and Transmission Channel Bandwidth

$d[k]$ effect

- For same NOISE or SNR, fewer $n = \log_2 L$ bits
- For same n, reduced noise or higher SNR



$d[k] = m[k] - m[k\text{-}1]$

decreases peak d_p

decreases Δv_d

decreases NOISE

PCM and Transmission Channel Bandwidth

At the receiver

- $m[k]$ is reconstructed from $d[k]$ and previous samples $m[k\text{-}1]$

$$m[k] = d[k] + m[k\text{-}1]$$

PCM and Transmission Channel Bandwidth

Findings

- $d[k]$ reduces the no. of transmitted bits
- Smaller is $d[k]$, so is n

PCM and Transmission Channel Bandwidth

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Effective prediction of $m[k]$ => further reduction of $d[k]$

PCM and Transmission Channel Bandwidth

Findings

- $d[k]$ reduces the no. of transmitted bits
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Effective prediction of $m[k]$ => further reduction of $d[k]$

Assuming, $\hat{m}[k]$ is the predicted (estimated) value of $m[k]$

√ Calculate $d[k]$ as $d[k] = m[k] - \hat{m}[k]$

X instead of $d[k] = m[k] - m[k-1]$

PCM and Transmission Channel Bandwidth

Better is the prediction, closer is $\hat{m}[k]$ to $m[k]$

$\Downarrow$

Smaller is $d[k]$

$\Downarrow$

Smaller is n

PCM and Transmission Channel Bandwidth

Steps

Predict $\hat{m}[k]$

$\Downarrow$

Calculate the difference, $d[k] = m[k] - \hat{m}[k]$

$\Downarrow$

Represent $d[k]$ in fewer bits and transmit

PCM and Transmission Channel Bandwidth

Differential pulse code modulation (DPCM)

Predict $\hat{m}[k]$

↓

Calculate the difference, $d[k] = m[k] - \hat{m}[k]$

↓

Represent $d[k]$ in fewer bits and transmit

PCM and Transmission Channel Bandwidth

Differential pulse code modulation (DPCM)

difference betn successive samples is a special case of DPCM

In DPCM: $d[k] = m[k] - \hat{m}[k]$

In simple difference: $d[k] = m[k] - m[k-1]$

which assumes $\hat{m}[k] = m[k-1]$